

Life-Span Dendritic and Spine Changes in Areas 10 and 18 of Human Cortex: A Quantitative Golgi Study

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ABSTRACT

Dendritic neuropil is a sensitive indicator of the aging process and may exhibit regional cortical variations. The present study examined regional differences and age-related changes in the basilar dendrites/spines of supragranular pyramidal cells in human prefrontal (area 10) and secondary occipital (area 18) cortices. Tissue was obtained from the left hemisphere of 26 neurologically normal individuals ranging in age from 14 to 106 years ($M_{\text{age}} = 57 \pm 22$ years; 13 males, 13 females). In tissue prepared by a modified rapid Golgi technique, ten neurons were sampled from each cortical region ($N = 520$) and were evaluated according to the following parameters: total dendritic length, mean segment length, dendritic segment count, dendritic spine number, and dendritic spine density. The effects of age and Brodmann areas were analyzed with a nested multiple analysis of variance design.

Despite considerable interindividual variation, several clear findings emerged: 1) Dendritic systems were significantly larger in area 10 than in area 18 across the sampled life span, presumably because of the more integrative function of area 10 neurons. 2) There was a significant age effect, with a substantial decline in dendritic neuropil from the younger (≤ 50 years) group to the older (> 50 years) group, especially in spine measures, which decreased almost 50%. 3) Dendritic values were relatively stable after 40 years of age, suggesting that dendritic/spine degeneration in older, relatively healthy individuals may not be an inevitable consequence of the aging process. These findings underscore the importance of life-long commitment to a cognitively invigorating environment. *J. Comp. Neurol.* 386:661–680, 1997.

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Indexing terms: cerebral cortex; pyramidal cell; supragranular; basilar dendrite; aging

Normal aging in the mammalian brain is characterized by several interrelated metabolic and morphological changes, which mutually suggest that dendritic neuropil is a sensitive indicator of the aging process. Age-related metabolic alterations include a global reduction in the brain's energy requirements, as manifested by decreases in cerebral blood flow (Kety, 1956; Melamed et al., 1980; Shaw et al., 1984; Gur et al., 1987) and cerebral glucose utilization (Leenders et al., 1990; but cf. Yoshii et al., 1988). Given that up to 80% of glucose utilization is devoted to maintenance of the Na^+/K^+ ion exchange across cell membranes (Sokoloff, 1977, 1979; Purves, 1994) and that dendrites may constitute over 90% of a neuron's receptive surface area (Schadé and Baxter, 1960; Torikai et al., 1996), it is not surprising that qualitative observations of age-related degeneration in dendritic/spine systems (Conel, 1939–1967; Scheibel and Scheibel, 1975, 1976; Scheibel et al.,

1977; Watanabe, 1981) support an inverse relationship between brain glucose utilization and cerebral atrophy (Schlageter et al., 1987). Indeed, positron emission tomography (PET) studies (Chugani et al., 1987; Jacobs et al., 1995) and quantitative neuromorphological investigations (Huttenlocher, 1979, 1990; Rakic et al., 1986; Huttenlocher and de Courten, 1987; Jacobs and Scheibel, 1993) indicate that the aging pattern of cerebral metabolism closely tracks life-span histological changes. The present

Grant sponsor: National Science Foundation; Grant number: DUE-9550790; Grant sponsors: Hughes Foundation, John D. and Catherine T. MacArthur Foundation, The Colorado College.

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Received 7 January 1997; Revised 12 May 1997; Accepted 12 May 1997

Golgi analysis extends previous morphological research (Buell and Coleman, 1979, 1981; Flood et al., 1985; Nakamura et al., 1985; Jacobs and Scheibel, 1993; Anderson and Rutledge, 1996) by quantitatively exploring age-related dendritic/spine changes in left frontopolar (Brodmann's area 10) and occipital (Brodmann's area 18) regions of the human neocortex (Brodmann, 1909).

These particular areas were chosen for several reasons. First, neuroimaging studies suggest that different cortical areas in primates do not age in a uniform manner (Azari et al., 1992; Eberling et al., 1995). Poly- and supramodal cortical association areas, including the prefrontal cortex, appear particularly susceptible to aging, with relative sparing of sensory regions, such as the occipital cortex (Kuhl et al., 1982; Smith, 1984; Raz et al., 1993, 1997). Second, despite considerable research on the histological consequences (e.g., decreases in cell number) of aging in humans (Anderson et al., 1983; Terry et al., 1987), few quantitative dendritic investigations have addressed regional specificity in human aging because each study has generally focused on only one cortical area (see, e.g., Flood et al., 1985; Nakamura et al., 1985; Jacobs and Scheibel, 1993; Anderson and Rutledge, 1996). Thus, the present study is the first to examine quantitative age-related changes in human dendritic/spine systems across both areas 18 and 10.

Area 18, which constitutes about 10% of the human neocortex (Van Essen and Gallant, 1994), is homotypical isocortex involved in early visual processing (Burkhalter and Van Essen, 1986; Gegenfurtner et al., 1996); therefore, it represents unimodal association cortex (Benson, 1994). Reciprocal connections from layers 2, 3, and 4B of area 17 terminate in discrete thick, thin, and pale cytochrome oxidase (CO) stripes of area 18 (Livingstone and Hubel, 1984, 1987; Levitt et al., 1994), and input from the pulvinar corresponds precisely with the CO-rich zones (Levitt et al., 1995). These interconnections suggest that area 18 analyzes the retinal image entering the cortex (Stone et al., 1979), distributing functionally unique streams of information (e.g., color, form, orientation, binocular disparity, motion) to other extrastriate areas (Zeki, 1978; Livingstone and Hubel, 1987; Van Essen and Gallant, 1994). Although visual streams in area 18 subserve a segregative function (Levitt et al., 1994), many area 18 neurons are polyfunctionally selective for more than one parameter of visual information (Burkhalter and Van Essen, 1986; Gegenfurtner et al., 1996). Moreover, individual striate and extrastriate visual streams participate in crosstalk at different stages of processing (Felleman and Van Essen, 1991) and share thalamic afferents from the pulvinar (Levitt et al., 1995). Nevertheless, most information passing through area 18 retains anatomical and functional specificity as it is relayed to different regions of association cortex (DeYoe and Van Essen, 1985). Consequently, neurons in this region are concerned primarily with the processing of functionally grouped, unimodal information.

In contrast, the prefrontal cortex (including area 10) accounts for 29% of the human neocortex (Brodmann, 1912), represents homotypical associational isocortex, and constitutes the quintessential supramodal region of the brain (Benson, 1993, 1994). Thus, activation of the dorsolateral prefrontal area appears to be independent of sensory input and motor output; moreover, the more polar regions sampled in the present study are not consistently coactivated with other cortical areas (Melamed and Larsen,

1979; Roland, 1984, 1993). Tasks that activate the dorsolateral prefrontal region (e.g., delayed matching, verb generation, mental calculations) generally contain sequential contingencies and require processing of several representations (i.e., internal and external) of information (McCarthy et al., 1993; Roland, 1993; Swartz et al., 1995). By cross-temporally coordinating remote sensory association areas (Roland, 1993) and motor regions (Fuster et al., 1982), the prefrontal area appears to be involved in several higher integrative functions, including sequencing of information, drive, executive (cognitive) control, future memory, planning, and self-awareness (Stuss and Benson, 1984, 1986; Ingvar, 1985; Benson, 1994). Prefrontal cortex, which typically exhibits higher metabolism and regional cerebral blood flow in the normal, resting state than other cortical areas, including area 18 (Roland, 1984), is therefore critical for the selection, planning, and temporal organization of cognitive processes (Fuster, 1989).

Insofar as dendritic extent and specificity appear to be inversely related (Ramón y Cajal, 1902; Seldon, 1985), the above functional differences between areas 18 and 10 may be reflected by associated differences in the basilar dendritic systems of supragranular pyramidal cells, which form the object of the present study. This neuronal population was chosen for study because pyramidal cells are the numerically dominant cortical neuron (Feldman, 1984; Peters and Kara, 1985). Moreover, the relatively late differentiating supragranular cells (Ramón y Cajal, 1906; Yakovlev, 1962; Marin-Padilla, 1970a,b; Sauer et al., 1983; Becker et al., 1984) are crucial integrators of corticocortical activity (Meyer, 1987) and appear to be particularly sensitive to environmental influences (Diamond, 1967; Vaughan, 1977; Juraska, 1982). The basilar dendrites, in particular, develop well into adulthood (Buell and Coleman, 1981; Nakamura et al., 1985) and constitute a sensitive indicator of age-related alterations (Scheibel and Scheibel, 1975, 1976; Scheibel et al., 1977). Moreover, the integrity of these dendritic systems throughout the aging process is crucial to every aspect of cognitive activity (Scheibel, 1992).

Even more sensitive indicators of age-related changes are dendritic spines, which constitute the major loci of excitatory synaptic input to pyramidal neurons (for reviews, see Coss and Perkel, 1985; Horner, 1993; Harris and Kater, 1994) and serve as the basic functional unit of neuronal integration (Yuste and Denk, 1995). Spines are dynamic postsynaptic target structures that have been shown *in vitro* to change on a time scale of minutes (Dailey and Smith, 1996), with considerable plasticity exhibited in response to steroid hormones (Woolley et al., 1990; Murphy and Segal, 1996), social contact (Rollenhagen and Bischof, 1994), deafferentation (Ryugo et al., 1975), environmental interaction (Coss et al., 1980; Brandon and Coss, 1982), and excessive synaptic activity (Annis et al., 1994; Papa and Segal, 1996); as such, they are the prime morphological candidates for long-term neuronal plasticity and learning (Moser et al., 1994; Kawaguchi et al., 1995). With regard to age-related changes in spines, qualitative observations of spine reductions have been made in the aging human cortex (Scheibel et al., 1975), which are consistent with reports of age-related synaptic decreases (Huttenlocher, 1979). Of the few quantitative studies that exist, almost all sampled only limited portions of dendritic arbors. These investigations have revealed substantial losses in area 17 of mice (Leuba, 1983) and rats

(Feldman and Dowd, 1975) and in the prefrontal cortex of nonhuman primates (Uemura, 1980). To the best of our knowledge, there are only two quantitative studies of age-related spine changes in humans. Schierhorn (1981) noted substantial decreases in dendritic spine density in four cortical areas between the 4th and 8th decades. More recently, Anderson and Rutledge (1996) explored spine number on the basilar dendrites of supragranular pyramidal cells in the posterior superior temporal gyrus and noted a significant decline in spine number between 21 and 71 years.

The present paper is thus the first to quantify dendritic spines across the entire basilar dendritic array in two areas of the aging human cortex. Based on the above evidence and on preliminary research (Jacobs and Scheibel, 1993; Larsen et al., 1994; Courns et al., 1996), two primary hypotheses have guided the present investigation. First, insofar as the higher metabolic demands and the more integrative functional nature of area 10 require greater dendritic extent than the more discrete processing of area 18, measures of dendritic neuropil are expected to be consistently higher in area 10 across the sampled life span. Second, significant age-related decreases in dendritic neuropil, especially in spine measures, are expected in both cortical areas.

MATERIALS AND METHODS

Subjects

Tissue was obtained from 26 neurologically normal subjects ($M_{age} = 57 \pm 22$ years; 13 males; $M_{age} = 57 \pm 21$ years; 13 females; $M_{age} = 56 \pm 23$ years) ranging in age from 14 to 106 years (see Table 1). Autolysis time (AT) averaged 11 ± 6 hours (range: 1–23 hours); all brains were immersion fixed in 10% neutral buffered formalin for an average of 27 ± 18 days prior to staining. Several sources generously provided tissue: 1) Dr. D. Bowerman of the El Paso County coroner's office (seven brains); 2) Dr. R. Sherwin of Penrose Hospital in Colorado Springs, Colorado (17 brains); and 3) Drs. E. Orsini and W. Tyson of Denver's Children's Hospital (two brains). All possible historical information (e.g., age, gender, cause of death, agonal period) on each subject was obtained from autopsy reports and medical records. Tissue was excluded if the AT was unknown or if there were any signs of trauma, cerebral edema, chronic illness with central nervous system involvement, and so forth. The research protocol was approved by The Colorado College Human Subjects Review Board (H94-004).

Tissue selection and processing

Two tissue blocks (1–2 cm along the long axis of the gyrus) respectively representing areas 18 and 10 were removed from the lateral surface of the left hemisphere (see Fig. 1). For area 18, the sample block was located approximately 1.5 cm superior to the inferior surface of the occipital lobe and 2 cm from the midline. For area 10, the sample block was located approximately 1.5 cm lateral to the midline and 4 cm superior to the orbitomedial surface. Subsequent examination of Nissl-stained sections confirmed the cytoarchitectonic characteristics of each region.

Tissue blocks were trimmed to 3–5 mm in anteroposterior thickness and processed by a modified rapid Golgi technique (Scheibel and Scheibel, 1978). All tissue was coded before sectioning to prevent experimenter bias;

TABLE 1. Subject Summary

Subject ¹	Body weight (in kg)	Autolysis time (in hours)	Cause of death	Occupation (education) ²
M14	47	14	Cardiac arrhythmia	High school student (HS)
F22	—	1	Mucopolysaccharidosis type VI	—
M23	82	12	Motor vehicle accident	—
F32	77	20	Coronary artery heart disease	Nutritionist (UNI)
M32	74	9	Asphyxia (suicide)	—
F34	88	14.5	Heroin overdose	Computer programmer (UNI)
F38	91	10.5	Pneumonia	Union worker (UNI)
F46	56	5	Cardiac arrest, diabetes	Bank auditor (HS)
F48	73	4	Myxoid leiomyosarcoma	—
M50	87	22	Motor vehicle accident	Truck driver
F51	75	6	Emphysema, antitrypsin deficiency	Real estate agent
F56	65	17	Failed prosthetic mitral valve	Homemaker
M56	87	6	Sepsis, liver cirrhosis	Retired military colonel (UNI)
M63	74	23	Respiratory arrest	Retired army aviator
F64	100	11	Myocardial infarction	Homemaker (HS)
M66	71	9	Cardiac arrest, pneumonia	Utility worker (HS)
M69.1	58	7	Prostate cancer	Maintenance worker
M69.2	80	20	Coronary artery heart disease	Prison inmate
F72	—	14	Myocardial infarction	—
M73	80	22.5	Pulmonary hemorrhage	Real estate agent (UNI)
M74	81	6	Bronchopneumonia	—
M76	84	5	Cardiac arrhythmia	Farmer
F78	69	8.5	Myocardial infarction	Homemaker
M81	48	11	Gastrointestinal hemorrhage	Retired pediatrician (UNI)
F81	—	2.5	Pneumonia	Homemaker
F106	—	12	Pulmonary embolism	Retired civil service (UNI)

¹Subjects are referred to by gender (M, male; F, female) and by age in years. For example, M14 refers to a 14-year-old male. Because two male subjects were 69 years old, they were differentiated as M69.1 and M69.2. All subjects were Caucasian except M69.2 (Hispanic) and F106 (African-American).

²HS, high school; UNI, university.

codes were not revealed until completion of all tracings. To be consistent with previous research (Jacobs and Scheibel, 1993), processed tissue was serially sectioned at 120 μ m with a vibratome, such that the preparation was vertical to the pial surface and perpendicular to the long axis of the gyrus. This section thickness provided maximal clarity (i.e., minimal precipitation with good contrast between stained elements and background) and minimal disturbance of dendritic systems. Adjacent tissue blocks were stained with a modified cresyl echt violet technique (Gridley, 1960), permitting cytoarchitectonic comparisons for routine control purposes and for measures of laminar depth, which are expressed as the mean of five sample measurements taken across the crown of the gyrus.

Cell-selection criteria and quantification of data

Ten relatively isolated supragranular pyramidal cells per tissue block (i.e., 20 cells per brain) were chosen randomly for analysis following previously established criteria (Jacobs and Scheibel, 1993; Jacobs et al., 1993a,b): 1) The apical dendrite was perpendicular to the pial surface and was at least 100 μ m in length to confirm cell orientation. 2) The soma was located centrally within the 120- μ m section depth to minimize cutting of distal dendritic branches. 3) At least three primary basilar dendritic shafts were present, each with at least two secondary branches and their consequent branch systems. 4) Neurons showed no obvious evidence of incomplete impregna-

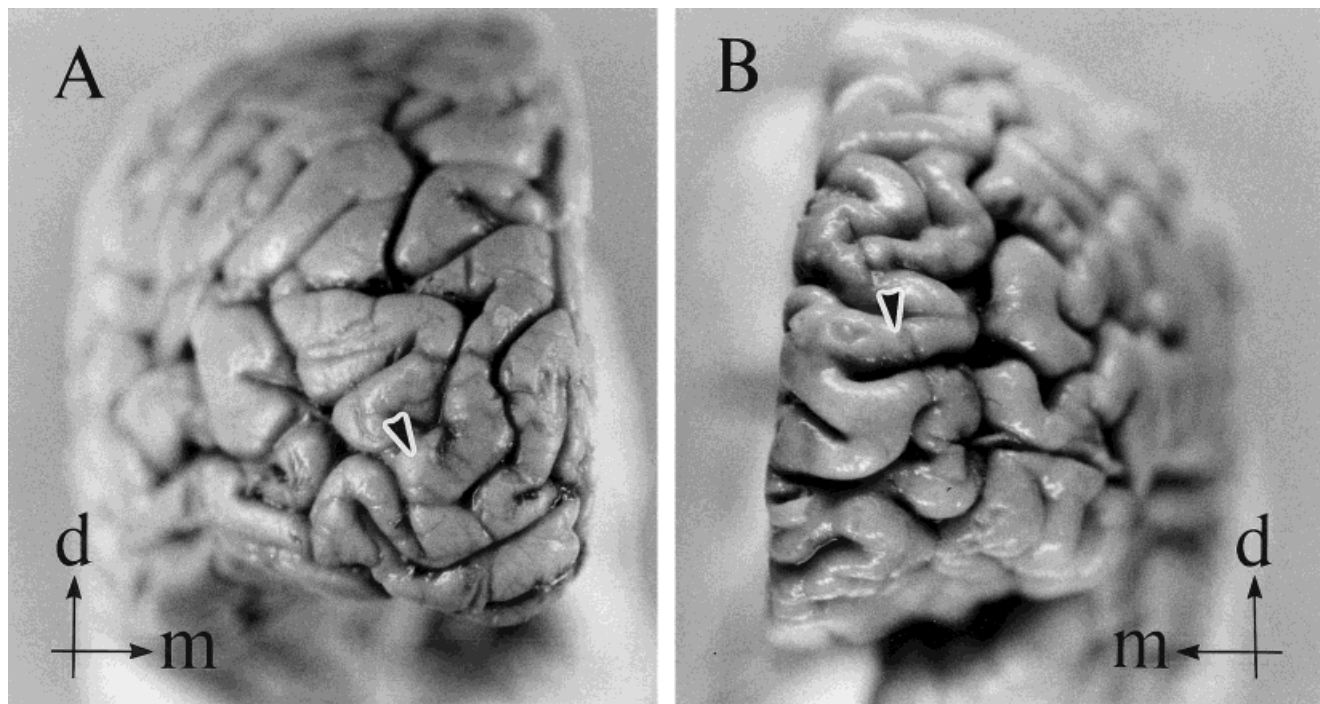


Fig. 1. Photographs of left hemisphere occipital (A) and frontal (B) poles of human brain illustrating relative position of tissue blocks in the present sample. Arrowheads indicate portion of area 18 analyzed from the occipital region and of area 10 from the prefrontal region. d, dorsal; m, medial.

tion (Williams et al., 1978). 5) Cells were relatively unobscured by adjacent neuronal structures. 6) Higher order branches should have natural terminations, characterized by either naturally tapered ends, growth cone-bearing tips, or terminal clusters of dendritic spines.

Because deeper lying neurons typically exhibit greater dendritic branching (Clendinnen and Eayrs, 1961; Volkmar and Greenough, 1972; McMullen et al., 1984; Matsubara et al., 1996) and layers II and III spanned approximately 1,000 μm in the present sample, a running average of soma depth was maintained as cells from each cortical area were drawn. Equating the average soma depth for each tissue block within each brain provided a rough control for soma size and helped to assure a more homogeneous cell population. Moreover, magnopyramidal neurons were not traced, nor was a distinction made between subtypes of pyramidal neurons (Matsubara et al., 1996). To further limit variability, all cells were sampled no farther than 1.5 cm from the tip of the gyral crown.

The first ten cells in each block meeting the above criteria were quantified on a Neurolucida system (Microbrightfield, Inc., Baltimore, MD) interfaced with an Olympus BH-2 microscope under a planachromat $\times 40$ (.70) dry objective. This system automatically calculated x -, y -, and z -coordinates to preserve the three-dimensional quality of the cells. Tracings proceeded in a specific sequence, beginning with the soma, which was traced at its widest point in the two-dimensional plane to provide a rough estimate of its cross-sectional area. After drawing the apical shaft, basilar dendrites were traced in their entirety along with all visible spines. No distinction was made between different spine types (e.g., stubby, mushroom shape, or thin; Horner, 1993). Dendritic processes were not followed into adjacent sections, nor was dendritic thickness examined.

Broken tips and unclear terminations were indicated as incomplete endings. Of the 30,763 dendritic segments quantified, 46% were intermediate segments. With regard to terminal segments, 46% were complete (area 18, 48%; area 10, 44%), and 54% were incomplete (area 18, 52%; area 10, 56%). Sectioned segments were not analyzed differentially because elimination of cells with incomplete segments would have biased the sample toward smaller neurons (Schadé and Caveness, 1968; Uylings et al., 1986).

Cells were traced by nine individuals. Intrarater reliability was determined by having each rater trace the same dendritic system (including somata and spines) ten times. There was little variation in tracings; the average coefficients of variation across all raters for soma size, total dendritic length, and dendritic spine number were 0.06, 0.04, and 0.05, respectively. To further test intrarater reliability, a split-plot design ($\alpha = 0.05$) compared the first five tracings with the second five tracings; no significant difference was found within raters for any of these measures. To maximize interrater reliability, all raters were normed before quantification. In tracings of ten different dendritic systems, Pearson product correlations across soma size, total dendritic length, and dendritic spine number averaged 0.94, 1.0, and 0.98, respectively. The tested agreement among raters was further evaluated by using an analysis of variance (ANOVA; $\alpha = 0.05$), which indicated no significant difference among raters on these measures. Finally, all tracings of neurons were reexamined by the primary investigator (B.J.) to assure quality control.

Dependent measures

Dendritic systems were quantified according to a centrifugal nomenclature (Bok, 1959; Van der Loos, 1959; Coleman and Riesen, 1968; Uylings et al., 1975, 1986):

Dendritic branches arising from the perikaryon are first-order segments until they bifurcate into second-order segments, which branch into third-order segments, and so on. Five measures characterized each cell's dendritic systems. Total dendritic length (TDL) refers to the summed length of dendritic segments. Mean segment length (MSL) represents the mean length of dendritic segments. Dendritic segment count (DSC) refers to the number of dendritic segments. Dendritic spine number (DSN) refers to the sum of all spines on dendritic segments. Dendritic spine density (DSD) represents the average number of spines per micron of dendritic length. A distinction was also made between the ontogenetically early developing proximal (first-, second-, and third-order) segments and the later developing distal (fourth-order and above) dendritic segments (Ramón y Cajal, 1909, 1911; Conel, 1939–1967). These dependent measures (represented as mean $\pm$ S.E.M.) are used to describe the entire dendritic system of a neuron, the proximal/distal segments, and/or individual segment orders for a particular independent variable.

Independent variables and statistical analyses

The raw data set was aggregated by neuron (CELL). Subsequent analyses evaluated the effects of age (AGE) and Brodmann areas (BRODMANN) on each of the five dependent measures by using a nested MANOVA design (SAS, Cary, NC; 6.08 for UNIX). In this model, BRODMANN was nested within AGE, and CELL was nested within AGE and BRODMANN. Because this design analyzed one dependent variable at a time, a Scheffé correction ($\alpha = 0.01$) was used to maintain an experimentwise α of 0.05.

To further understand the quantitative characteristics of the dendritic neuropil, subjects were placed into two age groups: a younger (≤ 50 years) group ($n = 10$; $M_{\text{age}} = 34 \pm 12$ years; $AT = 11 \pm 7$ hours) and an older (> 50 years) group ($n = 16$; $M_{\text{age}} = 71 \pm 13$ years; $AT = 11 \pm 6$ hours). This artificial age grouping is consistent with previous age-related studies (see, e.g., Jacobs and Scheibel, 1993) and helped to capture graphically substantial age-related changes.

RESULTS

Summary of neuronal sample

Several measures helped to minimize variability in the neuronal sampling procedure: 1) soma depth from pial surface, 2) soma size, and 3) laminar thickness. To explore further the relationship among these measures, two-tailed Pearson product-moment correlations based on the 520 sampled neurons were calculated (because of multiple correlations, a Scheffé correction was used to maintain an experimentwise α of 0.05; $\alpha = 0.001$).

Soma depth. The average soma depth for sampled neurons was very similar between age groups and the two Brodmann areas (Table 2) and placed the majority of somata in upper layer III. Soma size increased slightly but significantly with soma depth ($r[520] = 0.18$, $P < 0.0001$). There were also slight positive relationships between soma depth and TDL ($r[520] = 0.17$, $P < 0.0001$) and soma depth and DSC ($r[520] = 0.18$, $P < 0.0001$), further underscoring the necessity of controlling for soma depth.

Soma size. Most sampled neurons were small to medium-sized pyramidal cells, with the mean soma size

TABLE 2. Laminar and Sampled Soma Depths (μm) and Soma Size (μm^2)¹

	Younger group (≤ 50 years)		Older group (> 50 years)	
	Area 10	Area 18	Area 10	Area 18
Layer I/II junction	311 $\pm$ 44	258 $\pm$ 49	305 $\pm$ 41	257 $\pm$ 38
Layer II/III junction	593 $\pm$ 74	511 $\pm$ 71	555 $\pm$ 68	502 $\pm$ 100
Sampled soma depth (sampled soma size)	824 $\pm$ 253 (297 $\pm$ 9)	826 $\pm$ 245 (207 $\pm$ 7)	831 $\pm$ 196 (263 $\pm$ 8)	828 $\pm$ 170 (199 $\pm$ 6)
Layer III/IV junction	1,288 $\pm$ 57	1,321 $\pm$ 204	1,429 $\pm$ 163	1,289 $\pm$ 194
Gray/white matter junction	3,105 $\pm$ 343	2,783 $\pm$ 370	3,246 $\pm$ 517	2,807 $\pm$ 528

¹Values represent mean $\pm$ S.D.

larger in area 10 than in area 18 (Table 2). Although soma size appeared to decrease slightly from the younger group to the older group (Table 2), there was no significant correlation between age and soma size. As soma size increased, so did TDL ($r[520] = 0.47$, $P < 0.0001$), DSC ($r[520] = 0.46$, $P < 0.0001$), and DSN ($r[520] = 0.23$, $P < 0.001$), indicating that larger somata generally exhibited more dendritic neuropil. There were no significant changes in MSL or DSD with soma size.

Laminar thickness. Overall laminar and cortical thickness appeared to be roughly comparable between the two age groups and between the two Brodmann areas, with area 10 somewhat thicker than area 18 (Table 2). Laminar II and III were easily distinguishable. Layer II was relatively thin and consisted primarily of densely packed, small pyramidal cells. Layer III was characterized by medium-sized pyramidal cells in the superficial portion, with magnopyramidal neurons intermittently distributed in the deeper sublayer. Layers IV–VI were grouped together. The present laminar values are consistent with those of previous cytoarchitectonic findings (von Economo and Koskinas, 1925; Zilles et al., 1986).

AT and dendritic measures

Golgi-impregnated tissue did not exhibit the autolytic changes (e.g., irregular varicose enlargements, constriction of dendrites) described by Williams et al. (1978). Moreover, partial correlation coefficients removing the effect of age revealed no significant relationship between AT and any of the dependent measures.

Dendritic findings

Photomicrographs of selected Golgi preparations indicate the overall quality of the stain (Fig. 2). It should be noted that, although cellular degeneration was observed in subpopulations of neurons in older brains (particularly in case F106; Fig. 3), these cells were not traced because they did not meet the selection criteria. In general, despite considerable interindividual variability, the present results indicate significant age-related changes in dendritic neuropil and significant differences between areas 10 and 18 across the sampled life span. These differences are depicted in sample Neurolucida tracings from the younger (Fig. 4) and the older (Fig. 5) age groups. Within each age group, dendritic systems generally appear more complex in area 10 than in area 18; between age groups, dendritic complexity in younger individuals appears greater than in older individuals for both cortical areas. These differences are explored below for each dependent measure.

Dendritic length. There was a significant AGE difference for both TDL [$F(21,22) = 2.63$, $P < 0.001$] and MSL [$F(21,22) = 4.82$, $P < 0.0001$]. An overview of the TDL (Fig. 6A) and MSL (Fig. 7A) values for each individual indicates

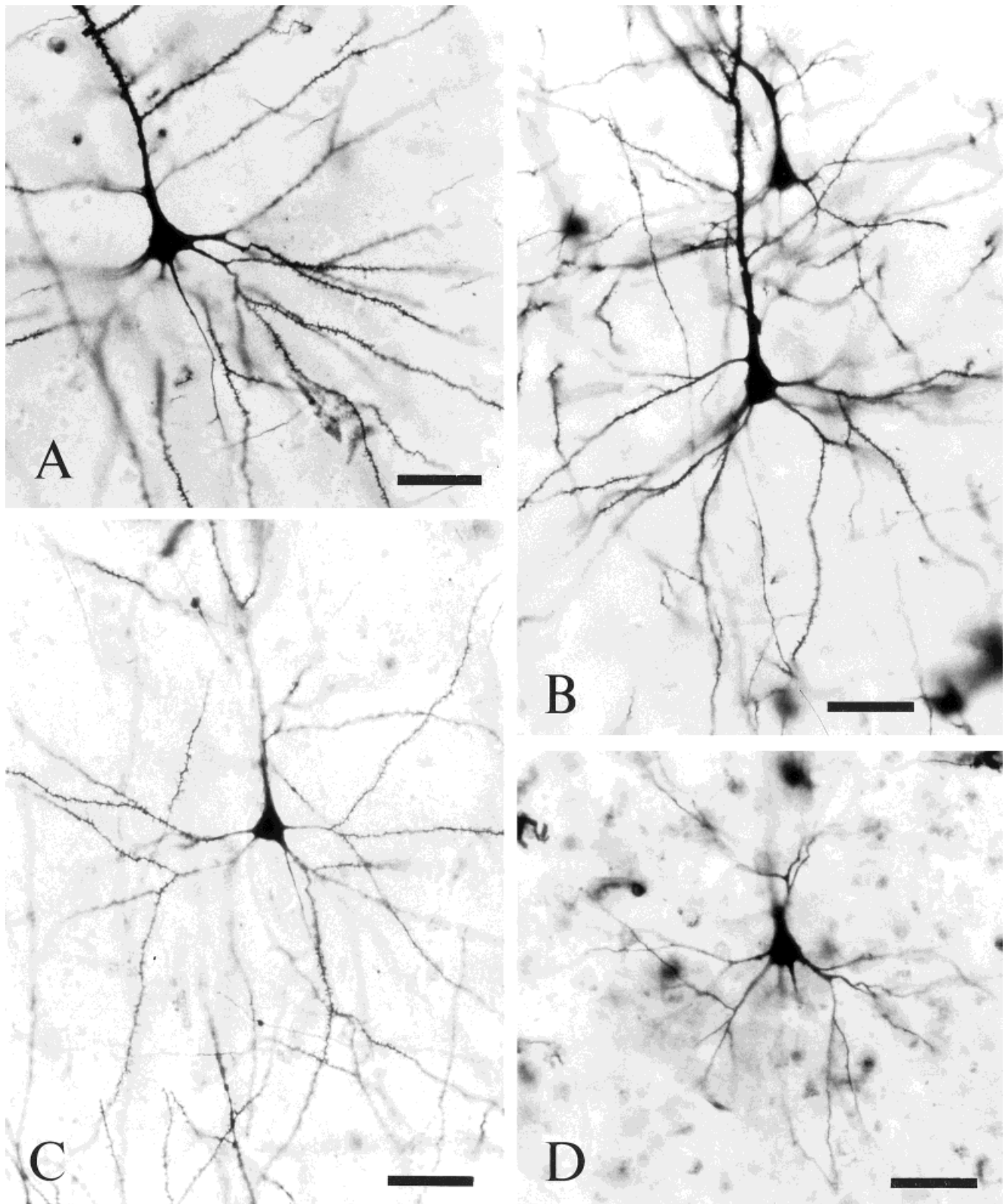


Fig. 2. Photomicrographs of area 10 supragranular pyramidal cells illustrating the sharp contour and intact quality of the Golgi-impregnated dendritic systems. Neurons from four cases are presented: a 14-year-old male (M14; **A**), M23 (**B**), a 32-year-old female (F32; **C**), and F106 (**D**). Scale bars = 50 μ m.

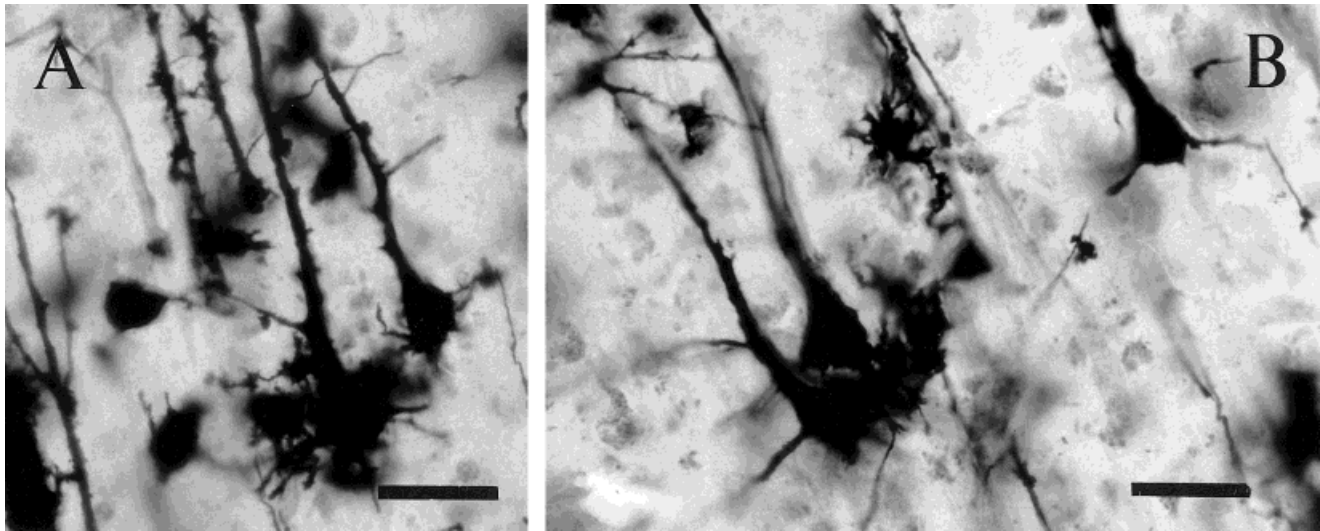


Fig. 3. Photomicrographs of degenerated neurons from F106. **A:** From area 18. **B:** From area 10. Note the virtually complete loss of basilar dendrites. Scale bars = 50 μ m.

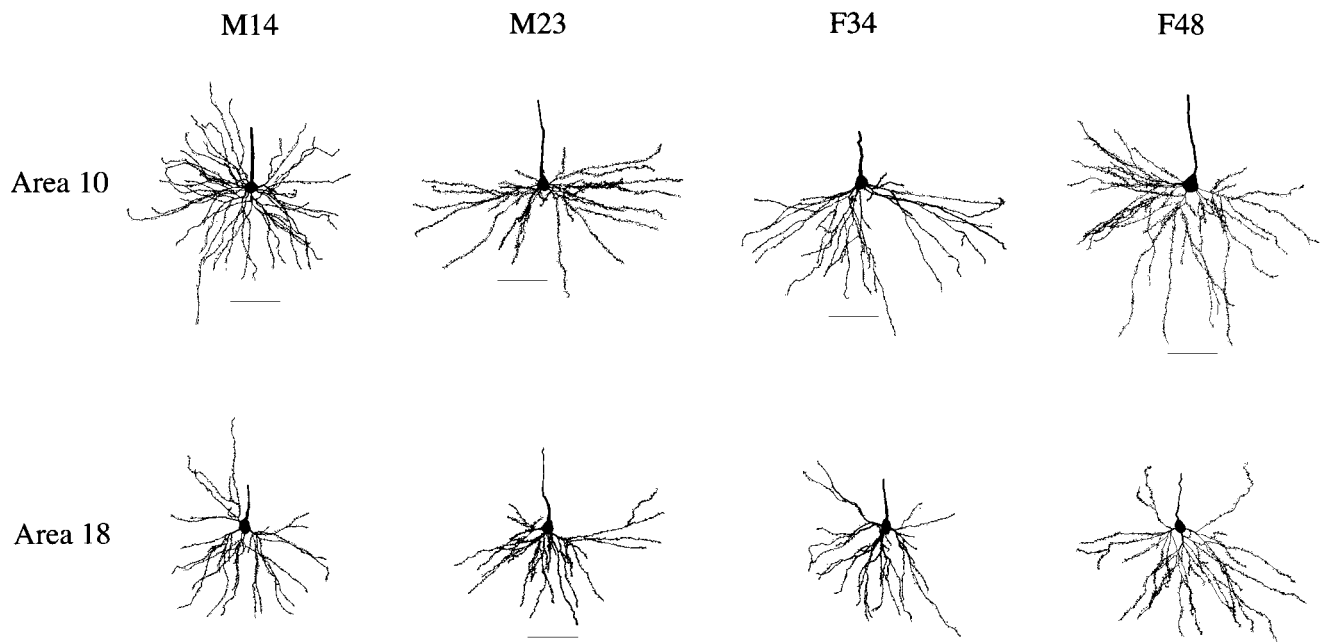


Fig. 4. Sample tracings of cells from both cortical areas in the younger group (≤ 50 years). On the whole, area 10 neurons appear somewhat more complex than area 18 neurons. Moreover, spine number/density are particularly high in M14 and M23 over F34 and F48. Scale bars = 100 μ m.

a slight decrease with age. This decrease was more apparent in comparisons of the younger and the older groups, where TDL decreased by 8.9% in area 10 and by 10.7% in area 18 (Fig. 6B). Similarly, MSL decreased from the younger group to the older group by 12.4% in area 10 and by 9.4% in area 18 (Fig. 7B). In general, distal segment length was substantially greater (TDL, 63.1%; MSL, 80.9%) than proximal segment length. Proximal segment TDL decreased an average of 8.0% from the younger group to the older group for both cortical areas, whereas distal TDL decreased an average of 10.8% (Fig. 6C). Comparable

decreases were observed in proximal (10.1%) and distal (11.6%) MSL measures (Fig. 7C). Examination of the dendritic envelope revealed that the middle of the dendritic tree (segment orders 3, 4, and 5) most clearly differentiated age groups and Brodmann areas (Fig. 6D). The MSL of each segment order increased in a somatofugal pattern, with consistent age-related decreases in the first five segment orders (Fig. 7D).

A significant BRODMANN nested within AGE effect obtained for TDL [$F(22,396) = 2.94$, $P < 0.0002$] but not MSL. TDL was higher in area 10 than in area 18 for all but

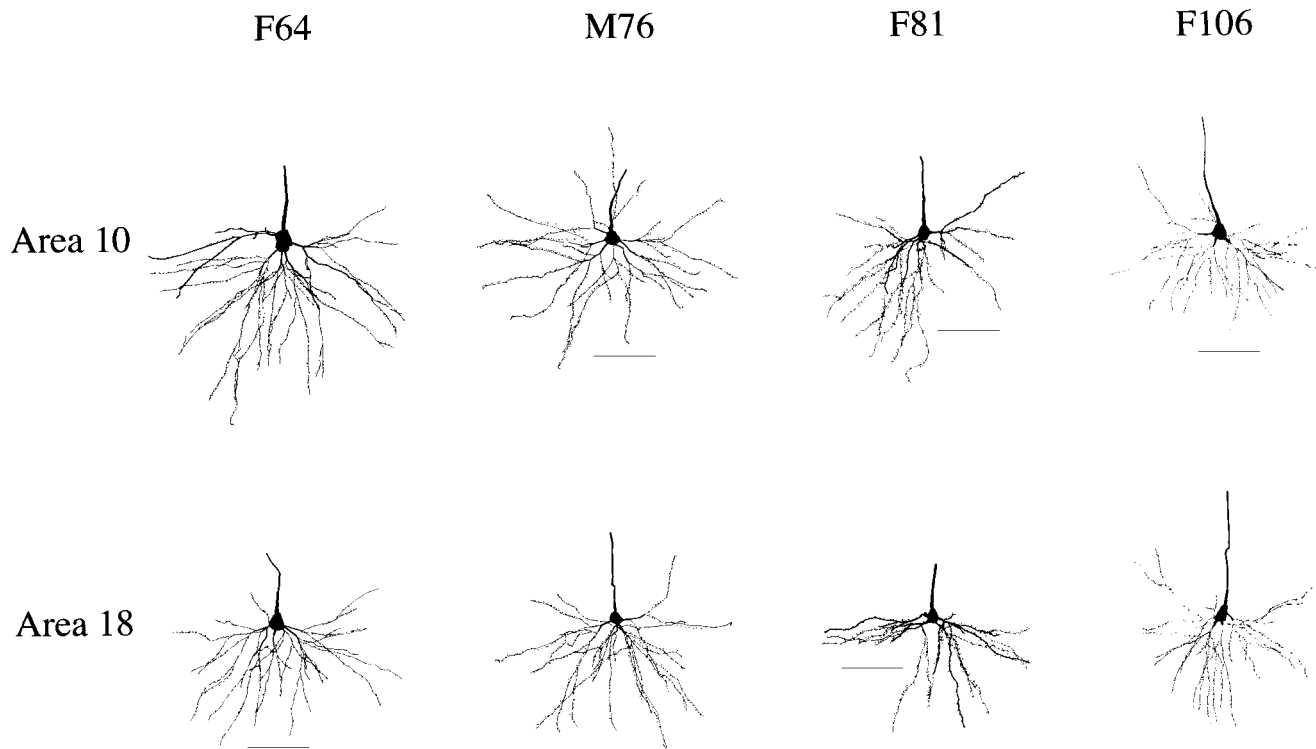


Fig. 5. Sample tracings of cells from both cortical areas in the older group (>50 years). Note the relative decrease in dendritic length and the significant loss of dendritic spines compared with cells in younger individuals (depicted in Fig. 4). Scale bars = 100 μ m.

two individuals (M73, M76), both of whom were in the older age group (Fig. 6A). In contrast, MSL values for area 10 were higher in only 15 out of the 26 cases (Fig. 7A). Although regional cortical differences were minimal within each age group for MSL (Fig. 7B), average area 10 TDL was 17.2% greater than area 18 TDL in the younger group and 19.6% greater in the older group (Fig. 6B). Both proximal and distal segments were longer in area 10 than in area 18 (Figs. 6C, 7C).

Dendritic number. There was a significant difference in DSC for AGE [$F(21,22) = 2.76$, $P < 0.0006$]. The number of segments fluctuated considerably from individual to individual (Fig. 8A). DSC increased by 4.0% from the younger group to the older group in area 10 and decreased by 2.5% in area 18 (Fig. 8B). Proximal segments outnumbered distal segments slightly in both age groups (Fig. 8C). The highest values, as with dendritic length, were exhibited by the middle of the dendritic envelope (especially segment orders 3 and 4), where clear differences emerged between age groups and Brodmann areas (Fig. 8D). Overall, the difference in DSC for BRODMANN nested within AGE was significant [$F(22,396) = 2.96$, $P < 0.0002$]. DSC values in area 10 were greater than those in area 18 for all but three cases (F32, M73, and M76; Fig. 8A). In the younger group, area 10 DSC was 11.4% higher than area 18 DSC; in the older group, this margin increased to 19.1% (Fig. 8B). DSC values were higher in area 10 than in area 18 for both proximal and distal segments (Fig. 8C).

Dendritic spines. The most substantial changes with age and between cortical areas emerged in dendritic spine measures. It is apparent in the photomicrographs (Fig. 9)

that there was a significant difference in AGE for both DSN [$F(21,22) = 24.57$, $P < 0.0001$] and DSD [$F(21,22) = 56.09$, $P < 0.0001$]. There was a marked decrease in DSN and DSD measures until the 4th decade, after which values fluctuated within a more constricted range (Figs. 10A, 11A). DSN and DSD values decreased by an average of 46% in both cortical areas from the younger group to the older group (Figs. 10B, 11B). In both the younger and the older age groups, DSN and DSD measures were substantially greater (an average of 96%) in distal over proximal segments across both cortical areas (Figs. 10C, 11C). DSN values were highest in the middle of the dendritic envelope, which also most clearly differentiated the age groups and Brodmann's areas (Fig. 10D). In contrast, DSD values increased somatofugally until the fourth-order segment, remaining at relatively high levels thereafter (Fig. 11D).

There was a significant BRODMANN nested within AGE effect for both DSN [$F(22,396) = 3.68$, $P < 0.0001$] and DSD [$F(22,396) = 2.57$, $P < 0.001$]. DSN measures were higher in area 10 over area 18 in all but three cases (M69.1, M73, and M76), all of whom were in the older age group (Fig. 10A). For DSD, area 10 values were greater than area 18 values in 19 of 26 cases, with 5 cases exhibiting the same DSD values between areas (Fig. 11A). Area 10 DSN was greater than area 18 DSN by 35.2% in the younger group and by 37.4% in the older group (Fig. 10B). These regional differences were somewhat less pronounced for DSD measures, where area 10 DSD was 14.3% greater than area 18 DSD in the younger group and 8.3% greater in the older group (Fig. 11B). Although area 10 consistently exhibited more spines than area 18, the

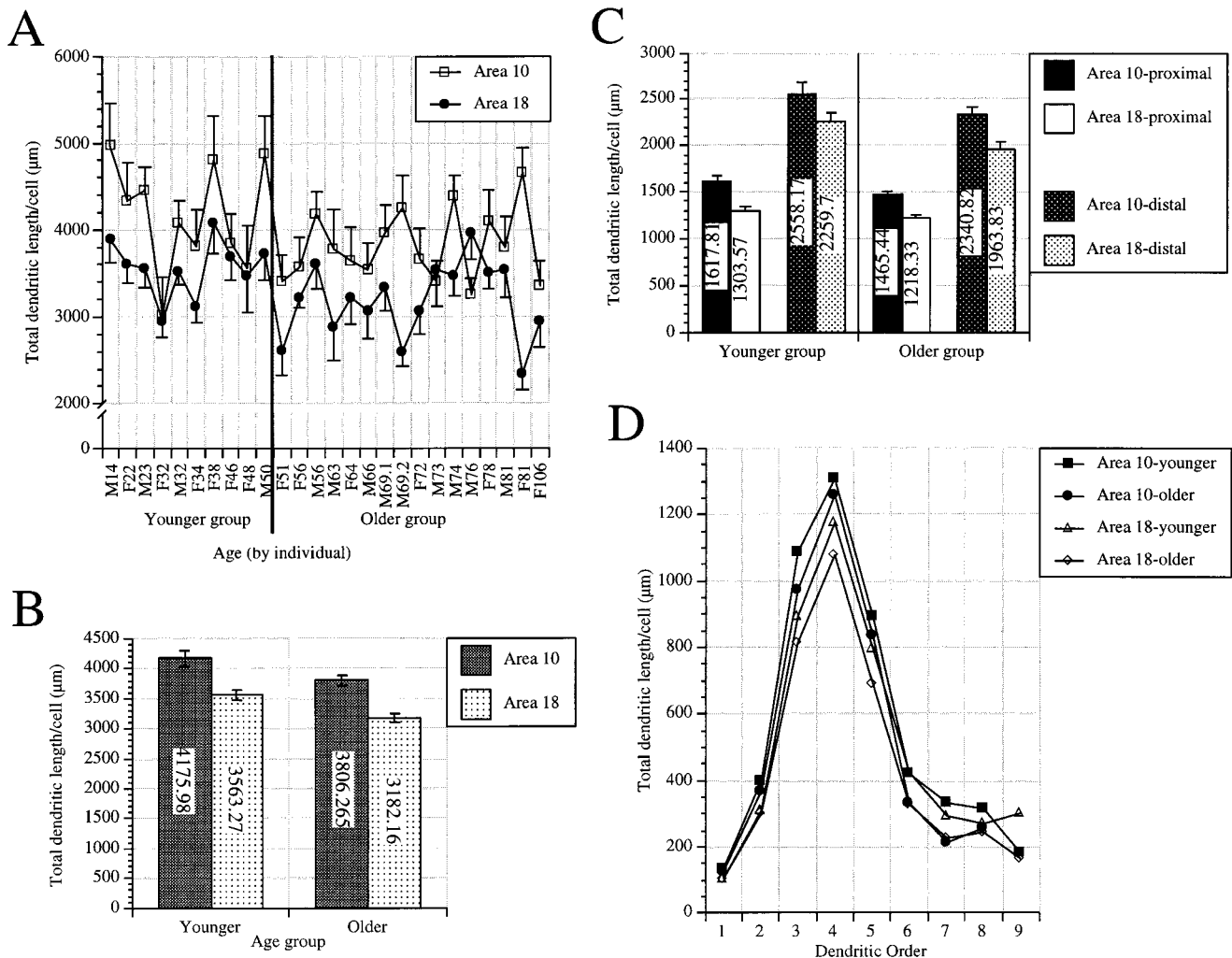


Fig. 6. Graphs of total dendritic length (TDL; in μm). **A**: Case-by-case breakdown of area 10 and area 18, with a division between individuals younger than or equal to 50 years of age and those individuals over 50 years of age. **B**: Bar graph illustrating area 10 and area 18 values for the younger and older age groups. **C**: Proximal and distal breakdowns for each Brodmann's area and each age group. **D**:

Dendritic envelopes for each level of Brodmann's area and each age group; dendritic branch orders 1 through 9 are represented along the abscissa. Note that, for all graphs, area 10 values are generally higher within a particular age group than area 18 values, and TDL values tended to decrease with age. Error bars represent S.E.M.

proximal-distal distribution of spines appeared similar between these two cortical areas (Figs. 10C, 11C).

DISCUSSION

Methodological considerations

The implications of the present results are tempered by several methodological factors (e.g., the practical limitations of human research, including retrospective analyses, small sample sizes, postmortem delay effects, consequences of hypoxia, cross-sectional vs. longitudinal designs), most of which have been addressed elsewhere (Williams et al., 1978; de Ruiter, 1983; Flood, 1993; Jacobs and Scheibel, 1993; Jacobs et al., 1993b) and will not be repeated in depth here. Nevertheless, several issues specific to the present study require elaboration, namely, 1) biological vs. chronological age, 2) Golgi stains vs. other histological techniques, 3) cell-selection criteria and sec-

tion thickness, and 4) light microscopic quantification of dendritic spines.

Biological vs. chronological age. A recognized limitation in human brain research is the extensive variability among individuals, including cortical gyral patterns between monozygotic twins (Bartley et al., 1997) and the relative lack of historical information about subjects (Scheibel, 1988; Scheibel et al., 1990; Jacobs et al., 1993b; Rajkowska and Goldman-Rakic, 1995). A closely related factor in aging studies is the discrepancy that may exist between chronological age and biological age (Ludwig and Smoke, 1980; Collier and Coleman, 1991). Insofar as individuals may age at different rates, chronologically older individuals in the present sample may not necessarily be biologically older. Therefore, within a restricted range, the difference between biological and chronological age may contribute to interindividual variation in the present investigation (especially within either the older or younger age groups). Minimizing the effects of such factors

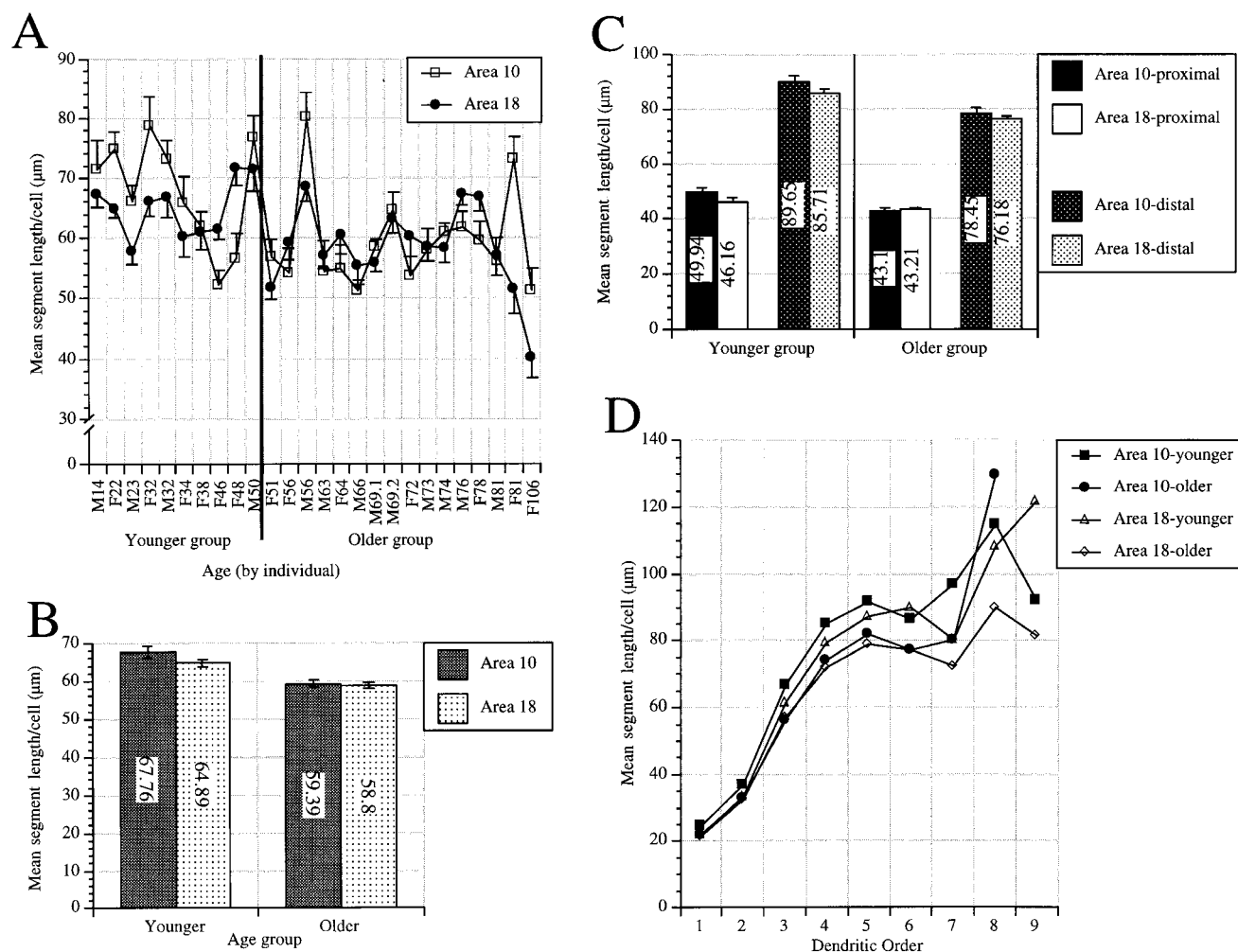


Fig. 7. Graphs of mean segment length (MSL; in μm). **A**: Case-by-case breakdown of area 10 and area 18, with a division between individuals younger than or equal to 50 years of age and those individuals over 50 years. **B**: Bar graph illustrating area 10 and area 18 values for the younger and older age groups. **C**: Proximal and distal breakdowns for each Brodmann's area and each age group. **D**:

Dendritic envelopes for each level of Brodmann's area and each age group; dendritic branch orders 1 through 9 are represented along the abscissa. Note that, for all graphs, area 10 values are generally higher within a particular age group than area 18 values, and MSL values tended to decrease with age. Error bars represent S.E.M.

requires a sufficiently large sample size to allow multiple subjects for each time point (Flood, 1993), which was not possible in the present investigation.

Golgi stains vs. other histological techniques. The relative merits of Golgi techniques are well documented (see, e.g., Scheibel and Scheibel, 1978; Buell, 1982; Braak and Braak, 1985; Meller and Dennis, 1990). Nevertheless, modern intracellular injection techniques have focused attention on the limitations of silver-impregnation methods. Although fluorescent dyes (e.g., Lucifer yellow) have proven effective in neuromorphological reconstructions in nonhuman animals (Trommald et al., 1995; Elston et al., 1996), they remain somewhat problematic in human autopsy tissue (Ohm and Diekmann, 1994; Belichenko and Dahlström, 1995) due to factors such as dye leakage in tissue with long postmortem and/or agonal periods. Moreover, injection techniques do not provide a random sampling of neurons, as appears to be the case with Golgi stains (Flood, 1993), and are less suited to large-scale quantitative studies, such as the present work. Therefore,

despite the development of new histological techniques, the Golgi method remains the stain of choice in extensive quantitative studies of immersion-fixed human tissue.

Cell-selection criteria and section thickness. Two neuronal populations appear to coexist in the aging human neocortex: one that is regressing and one that is proliferating (Buell and Coleman, 1979, 1981; Coleman and Buell, 1985; Flood et al., 1985). The present cell-selection criteria excluded severely regressing pyramidal cells from analysis (see Fig. 3), thereby skewing the sample in favor of neurons that appeared relatively healthy. Such a bias potentially attenuated the observed age-related decreases in dendritic neuropil, especially in older cortices, which contained a higher proportion of cells undergoing progressive dendritic regression (Scheibel, 1979, 1992). Thus, if a broader neuronal population had been sampled in the present study, then the age-related decreases documented may, in fact, have been greater than observed.

Sectioning tissue at 120 μm resulted in cut dendritic segments (Fig. 12). Therefore, measurements in the cur-

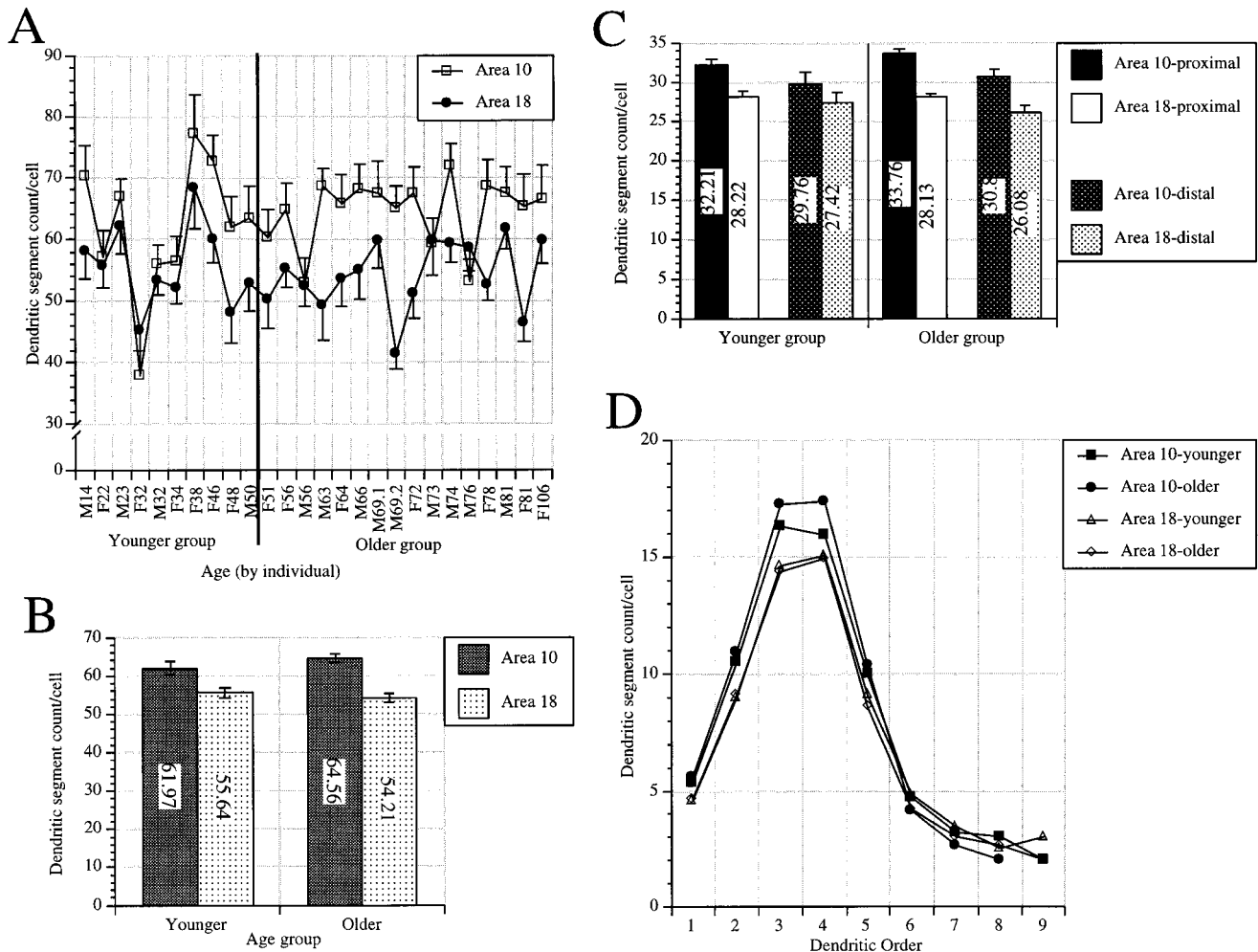


Fig. 8. Graphs of dendritic segment count (DSC). **A**: Case-by-case breakdown of area 10 and area 18 with a division between individuals younger than or equal to 50 years of age and those individuals over 50 years. **B**: Bar graph illustrating area 10 and area 18 values for the younger and older age groups. **C**: Proximal and distal breakdowns for each Brodmann's area and each age group. **D**: Dendritic envelopes for

each level of Brodmann's area and each age group; dendritic branch orders 1 through 9 are represented along the abscissa. Note that, for all graphs, area 10 values are generally higher within a particular age group than area 18 values, and DSC values remained relatively stable with age. Error bars represent S.E.M.

rent investigation represent relative rather than absolute values because they underestimate dendritic extent. To completely eliminate cut dendrites would require sections approximately 1,000 μm thick. Although serial section reconstructions of dendrites (Uylings et al., 1986) partially compensate for this underestimation, such a technique was not feasible in a study this extensive, where the entire basilar dendritic array was quantified. Therefore, to test the effects of sectioning, adjacent blocks of tissue from subject M32 were sectioned at 120 μm and 150 μm . Tracings of neurons at 150 μm ($n = 5$ per cortical area) were then compared with those traced in 120 μm sections ($n = 10$ per cortical area). In contrast to Koenderink and Uylings (1996), who found no significant difference in dendritic extent between varying section thicknesses with the Golgi-Cox method (presumably because of higher shrinkage effects compared with other Golgi techniques), the present rapid Golgi results indicate that TDL in 150- μm sections increased by 52% in area 10 and by 25% in area 18 over the 120- μm sections. Although proximal

segment TDL was not affected by the increased tissue thickness, distal TDL in 150- μm sections was 122% greater in area 10 and 50% greater in area 18 than in the 120- μm sections. The proportionally greater increase of dendritic neuropil in area 10 with increased section thickness suggests that the observed differences between area 10 and area 18 in the present study may actually underestimate the greater complexity of prefrontal over occipital dendritic systems.

Light microscopic quantification of dendritic spines. Because of the rapidity with which dendritic spines may change, the present study provides only a brief quantitative snapshot of spine number/density without reference to their heterogeneity. Moreover, because spines cannot be visualized directly above or below dendrites with light microscopy (as opposed to confocal laser-scanning microscopy), spine measures in the present study represent underestimates, especially for thicker dendrites (Horner and Arbuthnott, 1991). Although correction equations (Feldman and Peters, 1979) and three-dimensional recon-

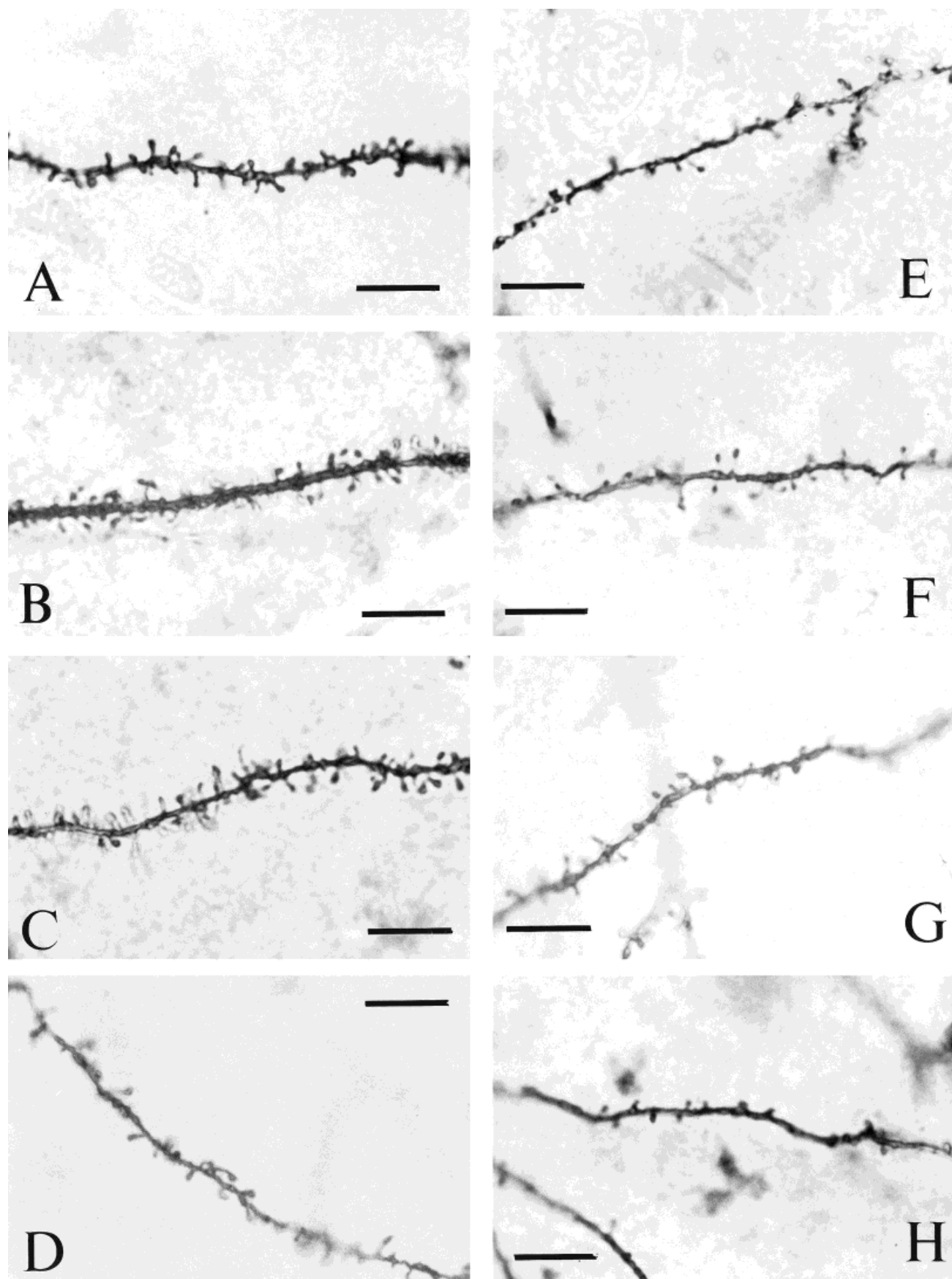


Fig. 9. Photomicrographs of spines in area 10 from eight subjects. Four subjects are younger than 50 years of age [M14 (A), M23 (B), F32 (C), and F48 (D)], and four subjects are older than 50 years of age [M56 (E), M73 (F), F81 (G), and F106 (H)]. Autolysis time ranged from 2.5

hours for F81 to 22.5 hours for M73. Note that, although dendritic spines are sharply contoured and relatively intact, there is a general decrease in spines with increasing age. Scale bars = 10 μ m.

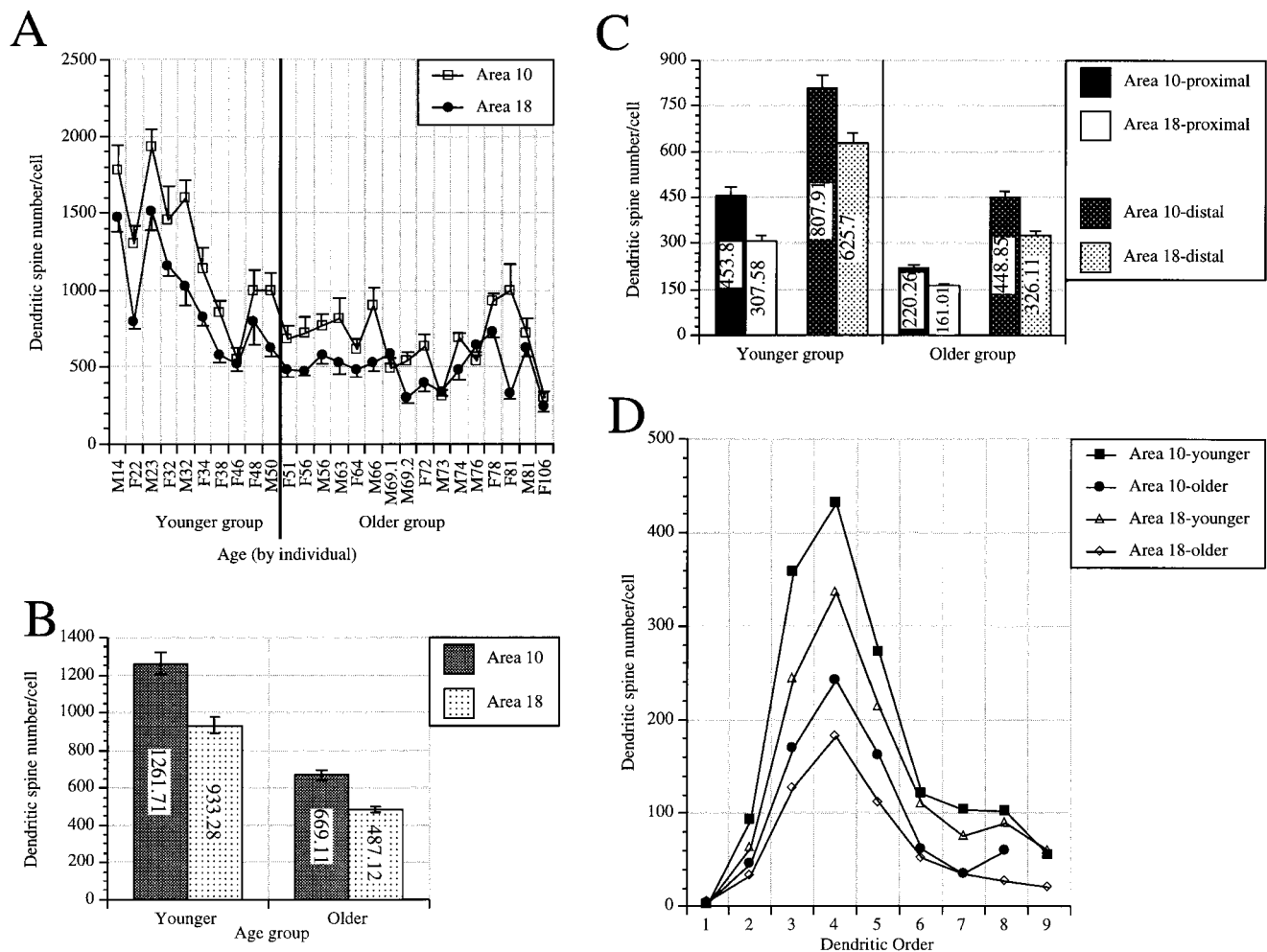


Fig. 10. Graphs of dendritic spine number (DSN). **A**: Case-by-case breakdown of area 10 and area 18, with a division between individuals younger than or equal to 50 years of age and those individuals over 50 years. **B**: Bar graph illustrating area 10 and area 18 values for the younger and older age groups. **C**: Proximal and distal breakdowns for each Brodmann's area and each age group. **D**: Dendritic envelopes for

each level of Brodmann's area and each age group; dendritic branch orders 1 through 9 are represented along the abscissa. Note that, for all graphs, area 10 values are generally higher within a particular age group than area 18 values. Moreover, DSN values decreased substantially before the age of 50 years and remained relatively stable thereafter. Error bars represent S.E.M.

structions of dendrites (Belichenko and Dahlström, 1995; Rusakov and Stewart, 1995) attempt to compensate for this underestimation, such techniques were not feasible in the present study because spines were quantified along the entire basilar dendritic array rather than along short (i.e., 50- μ m) sections. Importantly, the underestimation of spines is likely to be greatest in younger cortices, where spines are more numerous and may, in fact, obscure each other. Thus, the age-related decreases in spine measures observed in the present study may actually be greater than reported.

Regional differences in dendritic systems

Research on nonhuman animals has suggested the possibility of regional dendritic differences across cortical areas. For example, in both macaques and marmoset monkeys, the basilar dendritic field of layer III pyramidal cells appears to increase in a caudal-rostral direction, such that extrastriate areas exhibit larger fields than striate cortex (Lund et al., 1993; Elston et al., 1996). In humans, however, only preliminary investigations with limited

sample sizes have explored such regional variation. One series of studies documented greater dendritic complexity in the classical language regions of Broca and Wernicke than in sensorimotor cortices (Scheibel et al., 1985; Simonds and Scheibel, 1989; Jacobs et al., 1993a). Another pioneering investigation found that dendritic profiles in different cortical regions roughly reflected the computational tasks performed by those areas: dendritic arbors in two association cortices (areas 9 and 39) and in the finger region of the primary sensory strip were typically more complex than in the corresponding thoracic region (Scheibel et al., 1990). The present results extend the nonhuman research and preliminary human analyses by documenting, as predicted, that basilar dendrites of supragranular pyramidal cells in area 10 were significantly more extensive than those in area 18 for every measure except MSL. Regional differences were greatest for DSN, where spine numbers in area 10 were over 35% greater than in area 18. This finding is particularly significant because few, if any, investigations have ever documented regional spine differ-

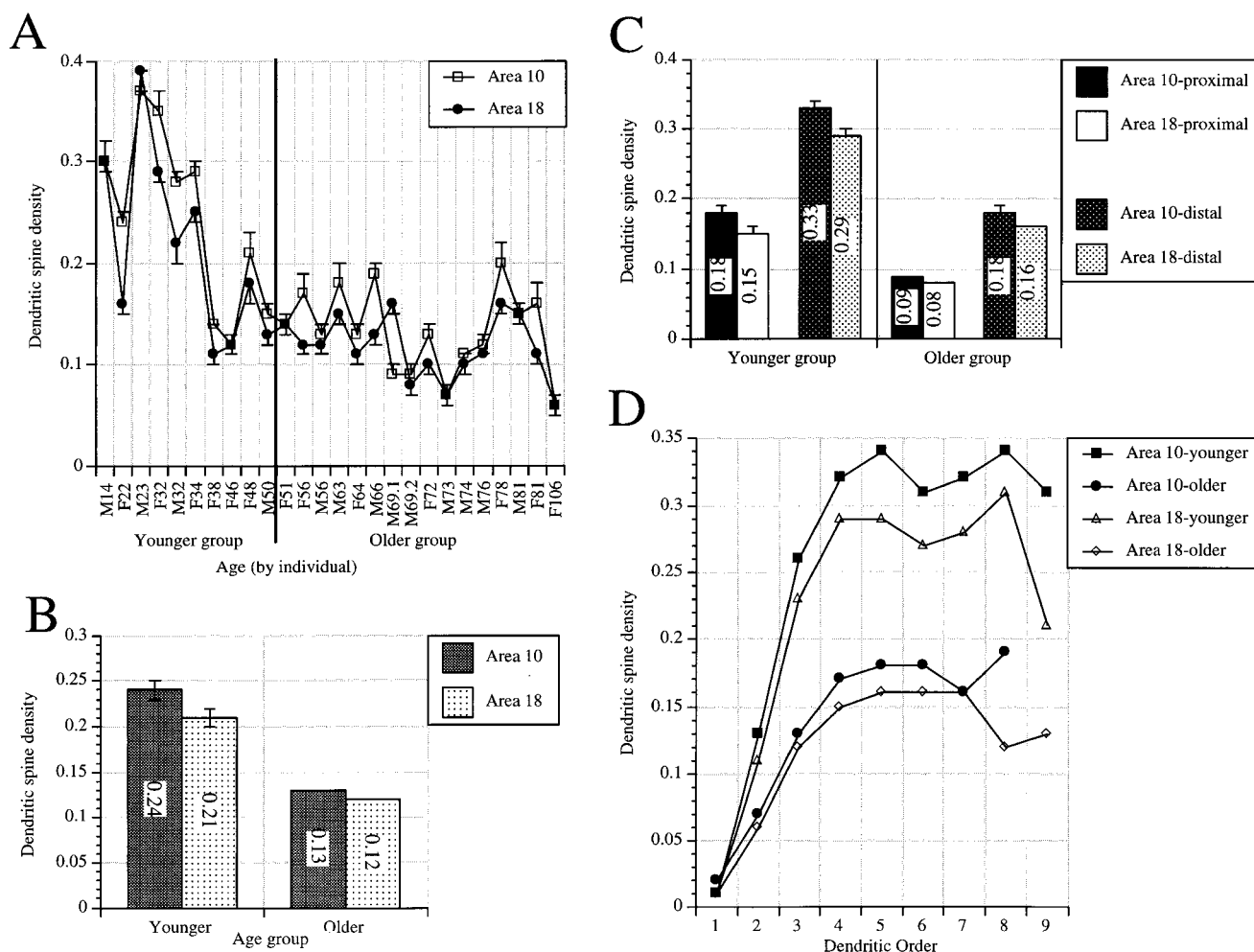


Fig. 11. Graphs of dendritic spine density (DSD). **A**: Case-by-case breakdown of area 10 and area 18, with a division between individuals younger than or equal to 50 years of age and those individuals over 50 years. **B**: Bar graph illustrating area 10 and area 18 values for the younger and older age groups. **C**: Proximal and distal breakdowns for each Brodmann's area and each age group. **D**: Dendritic envelopes for

each level of Brodmann's area and each age group; dendritic branch orders 1 through 9 are represented along the abscissa. Note that, for all graphs, area 10 values are generally higher within a particular age group than area 18 values. Moreover, DSD values decreased substantially before the age of 50 years and remained relatively stable thereafter. Error bars represent S.E.M.

ences in the cerebral cortex. Importantly, this regional advantage for area 10 was observed in almost all individual cases across the sampled life span. Exceptions to this pattern were primarily subjects in the older age group, indicating the possibility of selective age-related atrophy of the prefrontal region in these individuals, as has been suggested by previous aging research (Shaw et al., 1984; Coleman and Flood, 1987; Terry et al., 1987; Nielsen-Bohlman and Knight, 1995; Raz et al., 1997).

Although the functional significance of this regional dendritic advantage remains unclear, it is generally believed that the processing demands placed on dendritic systems substantially influence their ultimate expression (Ramón y Cajal, 1894; Hebb, 1949; Diamond et al., 1964; Scheibel et al., 1990; Jacobs et al., 1993b). In the present sample, the more complex dendritic array in area 10 neurons may facilitate a broader sampling of afferent information, thereby potentially increasing their integrative capacity. In contrast, the more limited basilar dendritic systems in area 18 neurons may correspond with

more discrete sampling of afferent information (i.e., smaller receptive fields; cf. Rosa and Schmid, 1995), as would be characteristic of information processing that is more unimodal than polymodal in nature. To the extent that the above suppositions are true, the present results appear to be consistent with the anatomical hierarchy proposed by Pandya and Yeterian (1990), and they provide initial support at the dendritic level for the functional hierarchy of cortical regions outlined by Benson (1993, 1994). According to this schema, area 18 contributes primarily to the initial stages of unimodal sensory processing whereas area 10 participates in supramodal monitoring of widespread, ongoing (sub)cortical activity. The present findings, as such, suggest that functional differences among various cortical areas can potentially be associated with concomitant variations in dendritic neuropil.

Life-span changes in dendritic systems

Age-related changes in both cortical regions followed the predicted pattern. In short, there were substantial de-

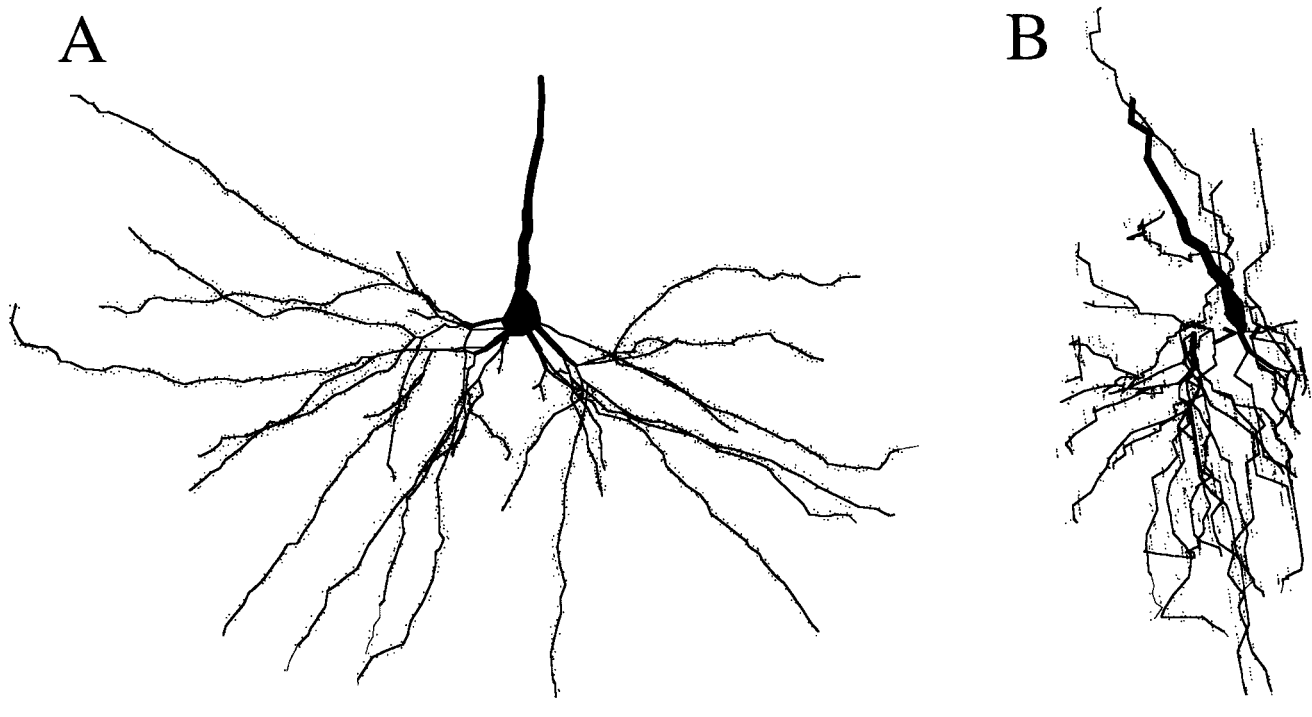


Fig. 12. Two views of one pyramidal cell from area 10 of case M32: one traced under the microscope (A) and one rotated 90° (B). In B, note the truncation of dendritic branches because of the 120- μ m tissue sections. Scale bars = 100 μ m.

creases in dendritic neuropil (especially in spine measures) by 40 years of age, with relative stability thereafter. These dendritic changes constitute only one of several life-span neural changes (Flood and Coleman, 1988; Scheibel, 1992; Peters et al., 1994), including associated declines in cerebral metabolism beyond 10 years of age (Melamed et al., 1980; Kuhl et al., 1982; Chugani et al., 1987; Leenders et al., 1990). Age-related changes in both dendrites and spines are considered below, followed by a brief discussion of environmental and individual factors.

Age-related dendritic changes. The overall decline in dendritic length reported in the present investigation provides further quantitative support for previous qualitative observations by the Scheibels (Scheibel and Scheibel, 1975, 1976; Scheibel et al., 1975), although these early observations partially confused normal aging with dementia-related degeneration (A.B. Scheibel, personal communication, April 28, 1997). Although methodological differences among studies make comparisons problematic (see Jacobs and Scheibel, 1993), the present results also appear to be generally consistent with previous quantitative findings (Nakamura et al., 1985; Anderson and Rutledge, 1996). In fact, the magnitude of dendritic regression in the present sample is nearly identical to that in a methodologically similar study on Wernicke's area (Jacobs and Scheibel, 1993): about an 11% decrease in both TDL and MSL from the younger to the older groups. These decreases were localized primarily to third-order and higher dendritic segments, thereby providing support for 1) the relative stability of more primary branches (Carughi et al., 1989), 2) the greater epigenetic sensitivity of more distal segments (Jones and Thomas, 1962; Smit et al.,

1972; Greenough and Volkmar, 1973; Uylings et al., 1978; Greenough et al., 1979; Davies and Katz, 1983), and 3) the distoproximal theory of dendritic degeneration (Feldman, 1977; Cupp and Uemura, 1980).

Although actual age-related increases in dendritic segment length could not be identified in the present investigation, previous research suggests that proliferative changes are likely (Buell and Coleman, 1979, 1981; Flood et al., 1985). Such length increases would partially compensate for dendritic regression, resulting in relative preservation of dendritic systems. Indeed, the moderate length decreases observed in the present study appear to be much less severe than in senile dementia (Yamada et al., 1988) and other types of neuropathology (Takashima et al., 1989; von Bossányi et al., 1992; Sotrel et al., 1993), where proliferative mechanisms may not be able to compensate as completely for degenerative processes (Scheibel and Tomiyasu, 1978; Paula-Barbosa et al., 1980; McKee et al., 1989; Flood, 1991). In addition, overall dendritic length regression in the present study was probably attenuated by the relative stability of DSC measures, which remained remarkably constant across the sampled life span. Although age-related dendritic retraction could theoretically "pull" more segments into the 120- μ m section, resulting in an apparent rather than actual increase in segment number, the cell-selection criteria in the present study minimized the likelihood of such an artifactual increase in DSC (see Jacobs and Scheibel, 1993).

Age-related spine changes. The present study provides a comprehensive quantitative account of dendritic spine number and density across the entire basilar dendritic array of human supragranular pyramidal cells.

These spine measures revealed the most dramatic age-related decrease in the current analysis: a nearly 50% loss of spines from the younger group to the older group. In discussing these results, it is essential to recognize that, apart from possible methodological (e.g., rapid Golgi vs. Golgi-Cox stains), species, and age influences, spine estimates in pyramidal cells differ across studies for a variety of reasons. First, spine measures vary greatly as a function of sampling distance from the soma (Globus and Scheibel, 1967; Marin-Padilla, 1967), in part because of differences in segment diameter (Bannister and Larkman, 1995). Second, although there is a high correlation ($r = 0.90$) between spine density on basilar dendrites and the apical shaft (Schüz, 1976), spine density is generally higher on apical than basilar dendrites (Globus and Scheibel, 1967; Uemura, 1980). Finally, it should be noted that sampling limited (i.e., 25–200 μm) sections of a dendrite, as is typically the case, provides a much less comprehensive spine estimate than does quantification of spines along the entire basilar dendritic array, as was done in the present study.

Accepting these factors, the present spine density estimates (ranging from 0.01 in first-order segments to 0.34 in fifth-order segments) are roughly comparable with values from other studies. With the rapid Golgi technique, Globus and Scheibel (1967) reported a basilar dendritic spine density of 0.3 in rabbit striate cortex. Uemura (1980) observed considerably higher densities (ranging from 0.61 to 0.86) on the basilar dendrites in monkey prefrontal cortex with a modified rapid Golgi stain. In normal adult human cortex, apical dendritic spine density has been estimated by means of a Golgi-Cox stain to be approximately 1.0 in layer III cells of area 17 (Michel and Garey, 1984). Estimates vary in other regions of human cortex: approximately 0.2–0.9 in layer III neurons of area 8 stained with a rapid Golgi method (Catalá et al., 1988) and 0.2–0.6 in cingulate gyrus and hippocampus stained with a modified Golgi technique (Mehraein et al., 1975).

Substantial age-related decreases in spine density have been reported in several species (for review, see Horner, 1993). In mice between 180 and 540 days of age, for example, an approximate loss of 50% was noted on the apical dendrites of layer III and V pyramidal cells with the Golgi-Cox method (Leuba, 1983). In the superior prefrontal gyrus of 7- to 28-year-old *Macaca mulatta*, dendritic spines on layer II and IV basilar dendrites decreased by 29% (Uemura, 1980). Although several human studies have examined age-related changes in synaptic density (Huttenlocher, 1979), few have specifically addressed spine density in the aging brain. By using a rapid Golgi technique, Anderson and Rutledge (1996) recently documented an inverse relationship ($r = -0.72$) between age and basilar dendritic spine number in ten males between 21 and 71 years of age. In the first study of its kind, Schierhorn (1981) quantified spine alterations in three prefrontal regions and the angular gyrus. In 20 individuals between 40 and 88 years of age, Schierhorn (1981) found with the Golgi-Cox technique that spine density (ranging from 0.35 to 0.45) on the apical dendrite of layer V neurons decreased by approximately 20–30%. In light of the restricted age range of Schierhorn's data, these spine decreases appear to be consistent with those observed in the present analysis.

Schierhorn (1981) suggested that these spine decreases were not necessarily caused by the aging process, per se,

but rather resulted from various types of damage (e.g., sensory deprivation); in particular, he underscored the importance of active interaction with the environment for healthy aging. In this regard, it is important to note that spine measures in the current study remained relatively stable after 40 years of age (see Figs. 10A, 11A) and appear to be consistent with previous quantitative findings (see, e.g., Anderson and Rutledge, 1996). In the absence of neurodegenerative diseases, which are associated with marked decreases in dendritic spines (Mehraein et al., 1975; Catalá et al., 1988), this relative stability suggests that substantial dendritic/spine losses may not be inevitable after the 4th decade in healthy, normal individuals.

Environmental and individual factors. Although several cognitive and behavioral consequences (e.g., changes in cognitive style, decreased processing speed) typically accompany the kinds of age-related structural alterations described in the current investigation (Harrington and Haaland, 1992; Grady et al., 1995), such consequences may not be inevitable (Rapp and Amaral, 1992) and are certainly subject to considerable individual variation. A critical factor in prolonging the relative stability of neuropil and the cognitive abilities it supports is continued neural activation (Swaab, 1991). Such activation appears to provide a redundancy of neuronal interconnectivity which, in turn, may serve as a buffer to age-related cognitive declines. In addition to the well-documented histological and behavioral benefits of enriched environments on nonhuman animals (see, e.g., Diamond et al., 1964, 1972; Connor and Diamond, 1982; Kiyono et al., 1985), recent research has demonstrated a positive association in humans between dendritic extent and education level (Jacobs et al., 1993b). A variety of studies also indicates that educational and occupational attainment, along with social contact, may provide a protective reserve capable of delaying cognitive decline (Wiederholt et al., 1993; Garfein and Herzog, 1995) as well as the clinical manifestation of Alzheimer's disease (Stern et al., 1994, 1995; Callahan et al., 1996).

Examining such factors requires detailed subject biographies, which were not available in the present investigation. Nevertheless, there was sufficient information to clarify some of the observed individual dendritic variation. For example, F22 had spine measures that were particularly low for her age (see Figs. 10A, 11A), especially compared with M23. This discrepancy may have been at least partially due to F22's prolonged illness (mucopolysaccharidosis type VI) and agonal period, whereas M23 was healthy at the time of his death from a motor vehicle accident. Case F32 also had dendritic values (TDL and DSC) that were particularly low (see Figs. 6A, 8A). Further research on this individual (unpublished data), who had a long history of bulimia, suggests that bulimia may have detrimental effects on distal dendritic segments (especially in prefrontal cortex), which, although it is only speculative, is at least consistent with the reported effects of malnutrition in nonhuman animals (Carughi et al., 1989). Finally, examination of individual spine measures (see Figs. 10A, 11A) revealed that, on the whole, cases M69.2–M76 had lower values than expected, especially given that spine measures were higher for older individuals (except for F106). One possible explanation is that very old individuals, in fact, may be healthier than individuals who are somewhat younger (Perls et al., 1993). Thus, cases F78–F81 (and perhaps even F106 in her 80s and 90s) may

have aged more robustly than cases M69.2–M76. Although these are speculative interpretations, they appear to be roughly consistent with current aging research. Only far more detailed background information on individual subjects will help clarify the relationship between environmental influence and dendritic measures.

CONCLUSIONS

The documented alterations in dendritic neuropil in the present study appear to be a natural extension of the dramatic developmental changes that mark the first 2 decades of the human life span. Despite considerable interindividual variation and the inherent limitations of a cross-sectional design, several clear findings emerged: 1) Dendritic systems were significantly greater in area 10 than in area 18 across the sampled life span, presumably because of the more integrative, supramodal function of area 10 neurons. More extensive research into regional variations in dendritic neuropil is currently in progress. 2) There was a substantial decline in dendritic neuropil from the younger group (≤ 50 years) to the older group (> 50 years), especially in spine measures, which decreased almost 50%. 3) Dendritic values were relatively stable after 40 years of age, suggesting that dendritic/spine degeneration in older, relatively healthy individuals may not be an inevitable consequence of the aging process. These findings indirectly support the notion that, for individuals to age robustly, they should maintain an involved, life-long commitment to a cognitively invigorating environment.

ACKNOWLEDGMENTS

Preliminary reports of some of these results have appeared in abstract form (Larsen et al., 1994; Courns et al., 1996). We gratefully acknowledge David Bowerman, Al Correl, Richard Sherwin, Leroy Fischer, Edmund Orsini, Wes Tyson, Kristi Erdal, and Arnold B. Scheibel for their generous assistance with this project. We also thank several students who participated in data collection: Marcy Wainwright, Kelly Courns, Becky Swanson, Michael Allen, Serapio Baca, Renee Ginardi, and Benjamin Addleson. Finally, we dedicate this work with admiration to the late Dr. Frank Benson.

LITERATURE CITED

- Anderson, B., and V. Rutledge (1996) Age and hemisphere effects on dendritic structure. *Brain* 119:1983–1990.
- Anderson, J.M., B.M. Hubbard, G.R. Coghill, and W. Slidders (1983) The effect of advanced old age on the neurone content of the cerebral cortex. *J. Neurol. Sci.* 58:233–244.
- Annis, C.M., D.K. O'Dowd, and R.T. Robertson (1994) Activity-dependent regulation of dendritic spine density on cortical pyramidal neurons in organotypic slice culture. *J. Neurobiol.* 25:1483–1493.
- Azari, N.P., S.I. Rapoport, J.A. Salerno, C.L. Grady, A. Gonzalez-Aviles, M.B. Schapiro, and B. Horwitz (1992) Interregional correlations of resting cerebral glucose metabolism in old and young women. *Brain Res.* 589:279–290.
- Bannister, N.J., and A.U. Larkman (1995) Dendritic morphology of CA1 pyramidal neurones from the rat hippocampus: II. Spine distributions. *J. Comp. Neurol.* 360:161–171.
- Bartley, A.J., D.W. Jones, and D.R. Weinberger (1997) Genetic variability of human brain size and cortical gyral patterns. *Brain* 120:257–269.
- Becker, L.E., D.L. Armstrong, F. Chan, and M.M. Wood (1984) Dendritic development in human occipital cortical neurons. *Dev. Brain Res.* 13:117–124.
- Belichenko, P.V., and A. Dahlström (1995) Studies of the 3-dimensional architecture of dendritic spines and varicosities in human cortex by confocal laser scanning microscopy and Lucifer Yellow microinjections. *J. Neurosci. Methods* 57:55–61.
- Benson, D.F. (1993) Prefrontal abilities. *Behav. Neurol.* 6:75–81.
- Benson, D.F. (1994) *The Neurology of Thinking*. New York: Oxford Press.
- Bok, S.T. (1959) *Histonomy of the Cerebral Cortex*. Amsterdam: Elsevier.
- Braak, H., and E. Braak (1985) Golgi preparations as a tool in neuropathology with particular reference to investigations of the human telencephalic cortex. *Progr. Neurobiol.* 25:93–139.
- Brandon, J.G., and R.G. Coss (1982) Rapid dendritic spine stem shortening during one-trial learning: The honeybee's first orientation flight. *Brain Res.* 252:51–61.
- Brodmann, K. (1909) *Vergleiche Lokalisation Lehre der Grosshirnrinde in ihren Prinzipien dargestellt auf Grund des Zellenbaues*. Leipzig: J.A. Barth.
- Brodmann, K. (1912) *Neue Ergebnisse über die vergleichende histologische Lokalisation der Grosshirnrinde mit besondere Berücksichtigung des Stirnhirns*. *Suppl. Anat. Anz.* 41:157–216.
- Buell, S.J. (1982) Golgi-Cox and rapid Golgi methods as applied to autopsied human brain tissue: Widely disparate results. *J. Neuropathol. Exp. Neurol.* 41:500–507.
- Buell, S.J., and P.D. Coleman (1979) Dendritic growth in aged human brain and failure of growth in senile dementia. *Science* 206:854–856.
- Buell, S.J., and P.D. Coleman (1981) Quantitative evidence for selective dendritic growth in normal human aging but not in senile dementia. *Brain Res.* 214:23–41.
- Burkhalter, A., and D.C. Van Essen (1986) Processing of color, form and disparity information in visual areas VP and V2 of ventral extrastriate cortex in the macaque monkey. *J. Neurosci.* 6:2327–2351.
- Callahan, C.M., K.S. Hall, S.L. Hui, B.S. Musick, F.W. Unverzagt, and H.C. Hendrie (1996) Relationship of age, education, and occupation with dementia among a community-based sample of African Americans. *Arch. Neurol.* 53:134–140.
- Carughi, A., K.J. Carpenter, and M.C. Diamond (1989) Effect of environmental enrichment during nutritional rehabilitation on body growth, blood parameters and cerebral cortical development of rats. *J. Nutr.* 119:2005–2016.
- Catalá, I., I. Ferrer, E., Galofré, and I. Fàbregues (1988) Decreased numbers of dendritic spines on cortical pyramidal neurons in dementia. A quantitative Golgi study on biopsy samples. *Hum. Neurobiol.* 6:255–259.
- Chugani, H.T., M.E. Phelps, and J.C. Mazziotta (1987) Positron emission tomography study of human brain functional development. *Ann. Neurol.* 22:487–497.
- Clendinnen, B.G., and J.T. Eayrs (1961) The anatomical and physiological effects of prenatally administered somatotrophin on cerebral development in rats. *J. Endocrinol.* 22:183–194.
- Coleman, P.D., and S.J. Buell (1985) Regulation of dendritic extent in developing and aging brain. In C.W. Cotman (ed): *Synaptic Plasticity*. New York: The Guilford Press, pp. 311–333.
- Coleman, P.D., and D.G. Flood (1987) Neuron numbers and dendritic extent in normal aging and Alzheimer's disease. *Neurobiol. Aging* 8:521–545.
- Coleman, P.D., and A.M. Riesen (1968) Environmental effects on cortical dendrite fields. I. Rearing in the dark. *J. Anat.* 102:363–374.
- Collier, T.J., and P.D. Coleman (1991) Divergence of biological and chronological aging: Evidence from rodent studies. *Neurobiol. Aging* 12:685–693.
- Conel, J. (1939–1967) *The Postnatal Development of the Human Cerebral Cortex*, Vols. 1–6. Cambridge, MA: Harvard University Press.
- Connor, J.R., and M.C. Diamond (1982) A comparison of dendritic spine number and type on pyramidal neurons of the visual cortex of old adult rats from social and isolated environments. *J. Comp. Neurol.* 210:99–106.
- Coss, R.G., and D.H. Perkel (1985) The function of dendritic spines: A review of theoretical issues. *Behav. Neural Biol.* 44:151–185.
- Coss, R.G., J.G. Brandon, and A. Globus (1980) Changes in morphology of dendritic spines on honeybee calycal interneurons associated with cumulative nursing and foraging experiences. *Brain Res.* 192:49–59.

- Courns, K., J. Ferguson, L. Larsen, M. Schall, and B. Jacobs (1996) Age-related dendritic and spine changes in human occipital and prefrontal cortices: A quantitative Golgi study. *Soc. Neurosci. Abstr.* 307:2.
- Cupp, C.J., and E. Uemura (1980) Age-related changes in prefrontal cortex of *Macaca mulatta*: Quantitative analysis of dendritic branching patterns. *Exp. Neurol.* 69:143–163.
- Dailey, M.E., and S.J. Smith (1996) The dynamics of dendritic structure in developing hippocampal slices. *J. Neurosci.* 16:2983–2994.
- Davies, C.A., and H.B. Katz (1983) The comparative effects of early-life undernutrition and subsequent differential environments on the dendritic branching of pyramidal cells in the rat visual cortex. *J. Comp. Neurol.* 218:345–350.
- de Ruiter, J.P. (1983) The influence of postmortem fixation delay on the reliability of the Golgi silver impregnation. *Brain Res.* 266:143–147.
- DeYoe, E.A., and D.C. Van Essen (1985) Segregation of efferent connections and receptive field properties in visual area V2 of the macaque. *Nature* 317:58–61.
- Diamond, M.C. (1967) Extensive cortical depth measurements and neuron size increases in the cortex of environmentally enriched rats. *J. Comp. Neurol.* 131:357–364.
- Diamond, M.C., D. Krech, and M.R. Rosenzweig (1964) The effects of an enriched environment on the histology of the rat cerebral cortex. *J. Comp. Neurol.* 123:111–119.
- Diamond, M.C., M.R. Rosenzweig, E.L. Bennett, B. Lindner, and L. Lyon (1972) Effects of environmental enrichment and impoverishment on rat cerebral cortex. *J. Neurobiol.* 3:47–64.
- Eberling, J.L., J.A. Roberts, D.J. De Manincor, K.M. Brennan, S.M. Hanrahan, H.F. Vanbrocklin, M.S. Roos, and W.J. Jagust (1995) PET studies of cerebral glucose metabolism in conscious rhesus monkeys. *Neurobiol. Aging* 16:825–832.
- Elston, G.N., M.G.P. Rosa, and M.B. Calford (1996) Comparison of dendritic fields of layer III pyramidal neurons in striate and extrastriate visual areas of the marmoset: A Lucifer yellow intracellular injection study. *Cerebral Cortex* 6:807–813.
- Feldman, M.L. (1977) Dendritic changes in the aging rat brain: Pyramidal cell dendrite length and ultrastructure. In K. Nandy and I. Sherwin (eds): *The Aging Brain and Senile Dementia*. New York: Plenum Press, pp. 23–27.
- Feldman, M.L. (1984) Morphology of the neocortical pyramidal neuron. In E.G. Jones and A. Peters (eds): *Cerebral Cortex*, Vol. 1. New York: Plenum Press, pp. 123–200.
- Feldman, M.L., and C. Dowd (1975) Loss of dendritic spines in aging cerebral cortex. *Anat. Embryol.* 148:279–301.
- Feldman, M.L., and A. Peters (1979) Technique for estimating total spine numbers on Golgi-impregnated dendrites. *J. Comp. Neurol.* 118:527–542.
- Felleman, D.J., and D.C. Van Essen (1991) Distributed hierarchical processing in the primate cerebral cortex. *Cerebral Cortex* 1:1–47.
- Flood, D.G. (1991) Region-specific stability of dendritic extent in normal human aging and regression in Alzheimer's disease. II. Subiculum. *Brain Res.* 540:83–95.
- Flood, D.G. (1993) Critical issues in the analysis of dendritic extent in aging humans, primates, and rodents. *Neurobiol. Aging* 14:649–654.
- Flood, D.G., and P.D. Coleman (1988) Neuron numbers and sizes in aging brain: Comparisons of human, monkey, and rodent data. *Neurobiol. Aging* 9:453–463.
- Flood, D.G., S.J. Buell, C.H. DeFlore, G.J. Horwitz, and P.D. Coleman (1985) Age-related dendritic growth in dentate gyrus of human brain is followed by regression in the "oldest old." *Brain Res.* 345:366–368.
- Fuster, J.M. (1989) *The Prefrontal Cortex: Anatomy, Physiology, and Neuropsychology of the Frontal Lobe*. New York: Raven Press.
- Fuster, J.M., R.H. Bauer, and J.P. Jervey (1982) Cellular discharge in the dorsolateral prefrontal cortex of the monkey in cognitive tasks. *Exp. Neurol.* 77:679–694.
- Garfein, A.J., and A.R. Herzog (1995) Robust aging among the young-old, old-old, and oldest-old. *J. Gerontol. B. Psychol. Sci. Soc. Sci.* 50:S77–S87.
- Gegenfurtner, K.R., D.C. Kiper, and S.B. Genstemaker (1996) Processing of color, form, and motion in macaque area V2. *Vis. Neurosci.* 13:161–172.
- Globus, A., and A.B. Scheibel (1967) Pattern and field in cortical structure: The rabbit. *J. Comp. Neurol.* 131:155–172.
- Grady, C.L., A.R. McIntosh, B. Horwitz, J.M. Maisog, L.G. Ungerleider, M.J. Mentis, P. Pietrini, M.B. Schapiro, and J.V. Haxby (1995) Age-related reductions in human recognition memory due to impaired encoding. *Science* 269:218–221.
- Greenough, W.T., and F.R. Volkmar (1973) Pattern of dendritic branching in occipital cortex of rats reared in complex environments. *Exp. Neurol.* 40:491–504.
- Greenough, W.T., J.M. Juraska, and F. Volkmar (1979) Maze training effects on dendritic branching in occipital cortex of adult rats. *Behav. Neural Biol.* 26:287–297.
- Gridley, M. (1960) *Manual of Histologic and Special Staining Techniques*, 2nd Ed. New York: McGraw-Hill.
- Gur, R.C., R.E. Gur, W.D. Obrist, B.E. Skolnick, and M. Reivich (1987) Age and regional cerebral blood flow at rest and during cognitive activity. *Arch. Gen. Psychiatr.* 44:617–621.
- Harrington, D.L., and K.Y. Haaland (1992) Skill learning in the elderly: Diminished implicit and explicit memory for a motor sequence. *Psychol. Aging* 7:425–434.
- Harris, K.M., and S.B. Kater (1994) Dendritic spines: Cellular specializations imparting both stability and flexibility to synaptic function. *Annu. Rev. Neurosci.* 17:341–371.
- Hebb, D. (1949) *The Organization of Behavior*. New York: John Wiley and Sons.
- Horner, C.H. (1993) Plasticity of the dendritic spine. *Progr. Neurobiol.* 41:281–321.
- Horner, C.H., and E. Arbutnot (1991) Methods of estimation of spine density—Are spines evenly distributed throughout the dendritic field? *J. Anat.* 177:179–184.
- Huttenlocher, P.R. (1979) Synaptic density in human frontal cortex—Developmental changes and effects of aging. *Brain Res.* 163:195–205.
- Huttenlocher, P.R. (1990) Morphometric study of human cerebral cortex development. *Neuropsychologia* 28:517–527.
- Huttenlocher, P.R., and C. de Courten (1987) The development of synapses in the striate cortex of man. *Hum. Neurobiol.* 6:1–9.
- Ingvar, D.H. (1985) "Memory of the future": An essay on the temporal organization of conscious awareness. *Hum. Neurobiol.* 4:127–136.
- Jacobs, B., and A.B. Scheibel (1993) A quantitative dendritic analysis of Wernicke's area in humans. I. Lifespan changes. *J. Comp. Neurol.* 327:83–96.
- Jacobs, B., H.A. Batal, B. Lynch, G. Ojemann, L. Ojemann, and A.B. Scheibel (1993a) Quantitative dendritic and spine analyses of speech cortices: A case study. *Brain Lang.* 44:239–253.
- Jacobs, B., M. Schall, and A.B. Scheibel (1993b) A quantitative dendritic analysis of Wernicke's area in humans. II. Gender, hemispheric, and environmental factors. *J. Comp. Neurol.* 327:97–111.
- Jacobs, B., H.T. Chugani, V. Allada, S. Chen, M.E. Phelps, D.B. Pollack, and M.J. Raleigh (1995) Developmental changes in brain metabolism in sedated rhesus macaques and vervet monkeys revealed by positron emission tomography. *Cerebral Cortex* 5:222–233.
- Jones, W.H., and D.B. Thomas (1962) Changes in the dendritic organization of neurons in the cerebral cortex following deafferentation. *J. Anat.* 96:375–381.
- Juraska, J.M. (1982) The development of pyramidal neurons after eye opening in the visual cortex of hooded rats: A quantitative study. *J. Comp. Neurol.* 212:208–213.
- Kawaguchi, S., M. Kishikawa, M. Sakae, and Y. Nakane (1995) Age-related changes in basal dendrite and dendritic spine of hippocampal pyramidal neurons (CA1) among SAMP1TA/NGs—quantitative analysis by the rapid Golgi method. *Mech. Aging Dev.* 83:11–20.
- Kety, S.S. (1956) Human cerebral blood flow and oxygen consumption as related to aging. *J. Chronic Dis.* 3:478–486.
- Kiyono, S., M.L. Seo, M. Shibagaki, and M. Inouye (1985) Facilitative effects of maternal environmental enrichment on maze learning in rat offspring. *Physiol. Behav.* 34:431–435.
- Koenderink, M.J.T., and H.B.M. Uylings (1996) Morphometric dendritic field analysis of pyramidal neurons in the human prefrontal cortex: Relation to section thickness. *J. Neurosci. Methods* 64:115–122.
- Kuhl, D.E., E.J. Metter, W.H. Riege, and M.E. Phelps (1982) Effects of human aging on patterns of local cerebral glucose utilization determined by the [¹⁸F] fluorodeoxyglucose method. *J. Cerebral Blood Flow Metabol.* 2:163–171.
- Larsen, L., R.L. Swanson, M.L. Wainwright, and B. Jacobs (1994) Quantitative dendritic and spine analyses of human prefrontal and occipital cortices. *Soc. Neurosci. Abstr.* 584:13.
- Leenders, K.L., D. Perani, A.A. Lammertsma, J.D. Heather, P. Buckingham, M.J.R. Healy, J.M. Gibbs, R.J.S. Wise, J. Hatazawa, S. Herold, R.P. Beaney, D.J. Brooks, T. Spinks, C. Rhodes, R.S.J. Frackowiak, and T. Jones (1990) Cerebral blood flow, blood volume and oxygen utilization: Normal values and effects of age. *Brain* 113:27–47.

- Leuba, G. (1983) Aging of dendrites in the cerebral cortex of the mouse. *Neuropathol. Appl. Neurobiol.* 9:467–475.
- Levitt, J.B., T. Yoshioka, and J.S. Lund (1994) Intrinsic cortical connections in macaque visual area V2: Evidence for interaction between different functional streams. *J. Comp. Neurol.* 342:551–570.
- Levitt, J.B., T. Yoshioka, and J.S. Lund (1995) Connections between the pulvinar complex and cytochrome oxidase-defined compartments in visual area V2 of macaque monkey. *Exp. Brain Res.* 104:419–430.
- Livingstone, M.S., and D.H. Hubel (1984) Anatomy and physiology of a color system in the primate visual cortex. *J. Neurosci.* 4:309–356.
- Livingstone, M.S., and D.H. Hubel (1987) Connections between layer 4B of area 17 and the thick cytochrome oxidase stripes of area 18 in the squirrel monkey. *J. Neurosci.* 7:3371–3377.
- Ludwig, F.C., and M.E. Smoke (1980) The measurement of biological age. *Exp. Aging Res.* 6:497–522.
- Lund, J.S., T. Yoshioka, and J.B. Levitt (1993) Comparison of intrinsic connectivity in different areas of macaque monkey cerebral cortex. *Cerebral Cortex* 3:148–162.
- Marin-Padilla, M. (1967) Number and distribution of the apical dendritic spines of the layer V pyramidal cells in man. *J. Comp. Neurol.* 131:475–490.
- Marin-Padilla, M. (1970a) Prenatal and early postnatal ontogenesis of the human motor cortex: A Golgi study. I. The sequential development of the cortical layers. *Brain Res.* 23:167–183.
- Marin-Padilla, M. (1970b) Prenatal and early postnatal ontogenesis of the human motor cortex: A Golgi study. II. The basket-pyramidal system. *Brain Res.* 23:185–191.
- Matsubara, J.A., R. Chase, and M. Thejomayen (1996) Comparative morphology of three types of projection-identified pyramidal neurons in the superficial layers of cat visual cortex. *J. Comp. Neurol.* 366:93–108.
- McCarthy, G., A.M. Blamire, D.L. Rothman, R. Gruetter, and R.G. Shulman (1993) Echo-planar magnetic resonance imaging studies of frontal cortex activation during word generation in humans. *Proc. Natl. Acad. Sci. USA* 90:4952–4956.
- McKee, A.C., N.W. Kowall, and K.S. Kosik (1989) Microtubular reorganization and dendritic growth response in Alzheimer's disease. *Ann. Neurol.* 26:652–659.
- McMullen, N.T., E.M. Glaser, and M. Tagamets (1984) Morphometry of spine-free nonpyramidal neurons in rabbit auditory cortex. *J. Comp. Neurol.* 222:383–395.
- Mehraein, P., M. Yamada, and E. Tarnowska-Dziduszko (1975) Quantitative study on dendrites and dendritic spines in Alzheimer's disease and senile dementia. In G.W. Kreutzberg (ed): *Advances in Neurology*, Vol. 12. New York: Raven Press, pp. 453–458.
- Melamed, E., and B. Larsen (1979) Cortical activation pattern during saccadic eye movements in humans: Localization by focal cerebral blood flow increases. *Ann. Neurol.* 5:79–88.
- Melamed, E., S. Lavy, S. Bentin, G. Cooper, and Y. Rinot (1980) Reduction in regional cerebral blood flow during normal aging in man. *Stroke* 11:31–35.
- Meller, S.T., and B.J. Dennis (1990) A multiple Golgi analysis of the periaqueductal gray in the rabbit. *J. Comp. Neurol.* 302:66–86.
- Meyer, G. (1987) Forms and spatial arrangement of neurons in the primary motor cortex of man. *J. Comp. Neurol.* 262:402–428.
- Michel, A.E., and L.J. Garey (1984) The development of dendritic spines in the human visual cortex. *Hum. Neurobiol.* 3:223–227.
- Moser, M.-B., M. Trommald, and P. Andersen (1994) An increase in dendritic spine density on hippocampal CA1 pyramidal cells following spatial learning in adult rats suggest the formation of new synapses. *Proc. Natl. Acad. Sci. USA* 91:12673–12675.
- Murphy, D.D., and M. Segal (1996) Regulation of dendritic spine density in cultured rat hippocampal neurons by steroid hormones. *J. Neurosci.* 16:4059–4068.
- Nakamura, S., I. Akiguchi, M. Kameyama, and N. Mizuno (1985) Age-related changes of pyramidal cell basal dendrites in layers III and V of human motor cortex: A quantitative Golgi study. *Acta Neuropathol. (Berlin)* 65:281–284.
- Nielsen-Bohlman, L., and R.T. Knight (1995) Prefrontal alterations during memory processing in aging. *Cerebral Cortex* 5:541–549.
- Ohm, T.G., and S. Diekmann (1994) The use of Lucifer yellow and mini-ruby for intracellular staining in fixed brain tissue: Methodological considerations evaluated in rat and human autopsy brains. *J. Neurosci. Methods* 55:105–110.
- Pandya, D.N., and E.H. Yeterian (1990) Prefrontal cortex in relation to other cortical areas in rhesus monkey: Architecture and connections. *Progr. Brain Res.* 85:63–94.
- Papa, M., and M. Segal (1996) Morphological plasticity in dendritic spines of cultured hippocampal neurons. *Neuroscience* 71:1005–1011.
- Paula-Barbosa, M.M., R. Mota Cardoso, M.L. Guimaraes, and C. Cruz (1980) Dendritic degeneration and regrowth in the cerebral cortex of patients with Alzheimer's disease. *J. Neurol. Sci.* 45:129–134.
- Perls, T.T., J.N. Morris, W.L. Ooi, and L.A. Lipsitz (1993) The relationship between age, gender and cognitive performance in the very old: The effect of selective survival. *J. Am. Geriatr. Soc.* 41:1193–1201.
- Peters, A., and D.A. Kara (1985) The neuronal composition of area 17 of rat visual cortex. I. The pyramidal cells. *J. Comp. Neurol.* 234:218–241.
- Peters, A., D. Leahu, M.B. Moss, and K.J. McNally (1994) The effects of aging on area 46 of the frontal cortex of the rhesus monkey. *Cerebral Cortex* 4:621–635.
- Purves, D. (1994) *Neural Activity and the Growth of the Brain*. Cambridge: Cambridge University Press.
- Rajkowska, G., and P.S. Goldman-Rakic (1995) Cytoarchitectonic definition of prefrontal areas in the normal human cortex: II. Variability in locations of areas 9 and 46 and relationship to the Talairach coordinate system. *Cerebral Cortex* 5:323–337.
- Rakic, P., J.-P. Bourgeois, M.F. Eckenhoff, N. Zecevic, and P.S. Goldman-Rakic (1986) Concurrent overproduction of synapses in diverse regions of the primate cerebral cortex. *Science* 232:232–235.
- Ramón y Cajal, S. (1894) *Les Nouvelles Idées sur la Structure du Système Nerveux chez L'homme et chez les Vertébrés* [Trans. by L. Azoulay]. Paris: Reinwald.
- Ramón y Cajal, S. (1902) *Studien über die Hirnrinde des Menschen*, III. Leipzig: Barth.
- Ramón y Cajal, S. (1906) *Studien über die Hirnrinde des Menschen V: Vergleichende Strukturbeschreibung und Histogenese der Hirnrinde. Anatomisch-physiologische Betrachtungen über das Gehirn. Struktur der Nervenzellen des Gehirns* [Trans. by J.A. Bressler]. Leipzig: Barth.
- Ramón y Cajal, S. (1909, 1911) *Histologie du Système Nerveux de L'homme et des Vertébrés*, 2 Vols [Trans. by L. Azoulay]. Paris: Maloine.
- Rapp, P.R., and D.G. Amaral (1992) Individual differences in the cognitive and neurobiological consequences of normal aging. *Trends Neurosci.* 15:340–345.
- Raz, N., I.J. Torres, W.D. Spencer, and J.D. Acker (1993) Pathoclysis in aging human cerebral cortex: Evidence from in vivo MRI morphometry. *Psychobiology* 21:151–160.
- Raz, N., F.M. Gunning, D. Head, J.H. Dupuis, J. McQuain, S.D. Briggs, W.J. Loken, A.E. Thornton, and J.D. Acker (1997) Selective aging of the human cerebral cortex observed in vivo: Differential vulnerability of the prefrontal gray matter. *Cerebral Cortex* 7:268–282.
- Roland, P.E. (1984) Metabolic measurements of the working frontal cortex in man. *Trends Neurosci.* 7:430–435.
- Roland, P.E. (1993) *Brain Activation*. New York: John Wiley and Sons.
- Rollenhagen, A., and H.-J. Bischof (1994) Spine morphology of neurons in the avian forebrain is affected by rearing conditions. *Behav. Neural Biol.* 62:83–89.
- Rosa, M.G.P., and L.M. Schmid (1995) Visual areas in the dorsal and medial extrastriate cortices of the marmoset. *J. Comp. Neurol.* 359:272–299.
- Rusakov, D.A., and M.G. Stewart (1995) Quantification of dendritic spine populations using image analysis and a tilting dissector. *J. Neurosci. Methods* 60:11–21.
- Ryugo, D.K., R. Ryugo, A. Globus, and H.P. Killackey (1975) Increased spine density in auditory cortex following visual or somatic deafferentation. *Brain Res.* 90:143–146.
- Sauer, B., G. Kammradt, I. Krauthausen, H.-J. Kretschmann, H.W. Lange, and F. Wingert (1983) Qualitative and quantitative development of the visual cortex in man. *J. Comp. Neurol.* 214:441–450.
- Schadé, J.P., and C.F. Baxter (1960) Changes during growth in the volume and surface area of cortical neurons in the rabbit. *Exp. Neurol.* 2:158–178.
- Schadé, J.P., and W.F. Caveness (1968) IV. Alteration in dendritic organization. *Brain Res.* 7:59–86.
- Scheibel, A.B. (1979) Dendritic changes in senile and presenile dementias. In R. Katzman (ed): *Congenital and Acquired Cognitive Disorders*. New York: Raven Press, pp. 107–122.
- Scheibel, A.B. (1988) Dendritic correlates of human cortical function. *Arch. Ital. Biol.* 126:347–357.
- Scheibel, A.B. (1992) Structural changes in the aging brain. In J. Birren (ed): *Handbook of Mental Health and Aging*, 2nd Ed. New York: Academic Press, pp. 147–173.

- Scheibel, A.B., and U. Tomiyasu (1978) Dendritic sprouting in Alzheimer's presenile dementia. *Exp. Neurol.* 60:1–8.
- Scheibel, A.B., L.A. Paul, I. Fried, A.B. Forsythe, U. Tomiyasu, A. Wechsler, A. Kao, and J. Slotnick (1985) Dendritic organization of the anterior speech area. *Exp. Neurol.* 87:109–117.
- Scheibel, A.B., T. Conrad, S. Perdue, U. Tomiyasu, and A. Wechsler (1990) A quantitative study of dendrite complexity in selected areas of the human cerebral cortex. *Brain Cogn.* 12:85–101.
- Scheibel, M.E., and A.B. Scheibel (1975) Structural changes in the aging brain. In H. Brody, D. Harman, and J.M. Ord (eds): *Clinical, Morphological and Neurochemical Aspects of the Aging Nervous System*. New York: Raven Press, pp. 17–31.
- Scheibel, M.E., and A.B. Scheibel (1976) Differential changes in old and new cortices. In K. Nandy and I. Sherwin (eds): *The Aging Brain and Senile Dementia*. Advances in Behavioral Biology, Vol. 23. New York: Plenum Press, pp. 39–58.
- Scheibel, M.E., and A.B. Scheibel (1978) The methods of Golgi. In R.T. Robertson (ed): *Neuroanatomical Research Techniques*. New York: Academic Press, pp. 89–114.
- Scheibel, M.E., R.D. Lindsay, U. Tomiyasu, and A.B. Scheibel (1975) Progressive dendritic changes in aging human cortex. *Exp. Neurol.* 47:392–403.
- Scheibel, M.E., U. Tomiyasu, and A.B. Scheibel (1977) The aging human Betz cell. *Exp. Neurol.* 56:598–609.
- Schierhorn, H. (1981) Strukturwandel neokortikaler Pyramidenneurone des Menschen während des 5. bis 9. Dezeniums. *Psychiatr. Neurol. Med. Psychol.* 33:664–673.
- Schlageter, N.L., B. Horwitz, H. Creasey, R. Carson, R. Duara, G.W. Berg, and S.I. Rapoport (1987) Relation of measured brain glucose utilization and cerebral atrophy in man. *J. Neurol. Neurosurg. Psychiatr.* 50:779–785.
- Schüz, A. (1976) Pyramidal cells with different densities of dendritic spines in the cortex of the mouse. *Z. Naturforsch.* 31:319–323.
- Seldon, H.L. (1985) The anatomy of speech perception: Human auditory cortex. In A. Peters and E.G. Jones (eds): *Cerebral Cortex*, Vol. 4. New York: Plenum Press, pp. 273–327.
- Shaw, T.G., K.F. Mortel, J.S. Meyer, R.L. Rogers, J. Hardenberg, and M.M. Cutaia (1984) Cerebral blood flow changes in benign aging and cerebrovascular disease. *Neurology* 34:855–862.
- Simonds, R.J., and A. B. Scheibel (1989) The postnatal development of the motor speech area: A preliminary study. *Brain Lang.* 37:42–58.
- Smit, G.J., H.B.M. Uylings, and L. Veldmaat-Wansink (1972) The branching pattern in dendrites of cortical neurons. *Acta Morphol. Neerl.-Scand.* 9:253–274.
- Smith, C.B. (1984) Aging and changes in cerebral energy metabolism. *Trends Neurosci.* 7:203–208.
- Sokoloff, L. (1977) Relation between physiological function and energy metabolism in the central nervous system. *J. Neurochem.* 29:13–26.
- Sokoloff, L. (1979) Mapping of local cerebral functional activity by measurement of local cerebral glucose utilization with (¹⁴C) deoxyglucose. *Brain* 102:653–668.
- Sotrel, A., R.S. Williams, W.E. Kaufmann, and R.H. Myers (1993) Evidence for neuronal degeneration and dendritic plasticity in cortical pyramidal neurons of Huntington's disease: A quantitative Golgi study. *Neurology* 43:2088–2096.
- Stern, Y., B. Gurland, T.K. Tatemichi, M.X. Tang, D. Wilder, and R. Mayeux (1994) Influence of education and occupation on the incidence of Alzheimer's disease. *JAMA* 271:1004–1010.
- Stern, Y., G.E. Alexander, I. Prohovnik, L. Stricks, B. Link, M.C. Lennon, and R. Mayeux (1995) Relationship between lifetime occupation and parietal flow: Implications for a reserve against Alzheimer's disease pathology. *Neurology* 45:55–60.
- Stone, J., B. Dreher, and A. Leventhal (1979) Hierarchical and parallel mechanisms in the organization of visual cortex. *Brain Res.* 180:345–394.
- Stuss, D.T., and D.F. Benson (1984) Neuropsychological studies of the frontal lobes. *Psychol. Bull.* 95:3–28.
- Stuss, D.T., and D.F. Benson (1986) *The Frontal Lobes*. New York: Raven Press.
- Swaab, D.F. (1991) Brain aging and Alzheimer's disease, "wear and tear" vs. "use it or lose it." *Neurobiol. Aging* 12:317–324.
- Swartz, B.E., E. Halgren, J.M. Fuster, F. Simpkins, M. Gee, and M. Mandelkern (1995) Cortical metabolic activation in humans during a visual memory task. *Cerebral Cortex* 5:205–214.
- Takashima, S., A. Ieshima, H. Nakamura, and L.E. Becker (1989) Dendrites, dementia and the down syndrome. *Brain Dev.* 11:131–133.
- Terry, R.D., R. DeTeresa, and L.A. Hansen (1987) Neocortical cell counts in normal human adult aging. *Ann. Neurol.* 21:530–539.
- Torikai, H., F. Hayashi, K. Tanaka, T. Chiba, Y. Fukuda, and H. Moriya (1996) Recruitment order and dendritic morphology of rat phrenic motoneurons. *J. Comp. Neurol.* 366:231–243.
- Trommald, M., V. Jensen, and P. Andersen (1995) Analysis of dendritic spines in rat CA1 pyramidal cells intracellularly filled with a fluorescent dye. *J. Comp. Neurol.* 353:260–274.
- Uemura, E. (1980) Age-related changes in prefrontal cortex of *Macaca mulatta*: Synaptic density. *Exp. Neurol.* 69:164–172.
- Uylings, H.B.M., G.J. Smit, and W.A.M. Veltman (1975) Ordering methods in quantitative analysis of branching structures of dendritic trees. In G.W. Kreutzberg (ed): *Advances in Neurology*, Vol. 12. New York: Raven Press, pp. 247–254.
- Uylings, H.B.M., K. Kuypers, M.C. Diamond, and W.A.M. Veltman (1978) The effects of differential environments on plasticity of cortical pyramidal neurons in adult rats. *Exp. Neurol.* 62:658–677.
- Uylings, H.B.M., A. Ruiz-Marcos, and J. van Pelt (1986) The metric analysis of three-dimensional dendritic tree patterns: A methodological review. *J. Neurosci. Methods* 18:127–151.
- Van der Loos, H. (1959) *Dendro-dendritic Connections in the Cerebral Cortex* [in Dutch]. Doctoral dissertation, University of Amsterdam. Haarlem: Stam.
- Van Essen, D.C., and J.L. Gallant (1994) Neural mechanisms of form and motion processing in the primate visual system. *Neuron* 13:1–10.
- Vaughan, D.W. (1977) Age-related deterioration of pyramidal cell basal dendrites in rat auditory cortex. *J. Comp. Neurol.* 171:501–516.
- Volkmar, F.R., and W.T. Greenough (1972) Rearing complexity affects branching of dendrites in the visual cortex of the rat. *Science* 176:1445–1447.
- von Bossányi, V. P., J.A. Schmitt, and K. Dietzmann (1992) Dendriten-Morphometrie von Lamina-V-Pyramidenzellen bei frühkindlicher Hirnschädigung—eine Golgi-studie. *Acta Histochem.* 42:S267–S272.
- von Economo, C., and G.N. Koskinas (1925) *Die Cytoarchitektonik der Hirnrinde des erwachsenen Menschen*. Berlin: Springer.
- Watanabe, H. (1981) Aging changes of apical dendritic spines in the area 4 human pyramidal cells with the rapid Golgi method. *Clin. Neurol.* 21:895–902.
- Wiederholt, W.C., D. Cahn, N.M. Butters, D.P. Salmon, D. Kritz-Silverstein, and E. Barrett-Connor (1993) Effects of age, gender and education on selected neuropsychological tests in an elderly community cohort. *J. Am. Geriatr. Soc.* 41:639–647.
- Williams, R.S., R.J. Ferrante, and V.S. Caviness, Jr. (1978) The Golgi rapid method in clinical neuropathology: Morphological consequences of suboptimal fixation. *J. Neuropathol. Exp. Neurol.* 37:13–33.
- Woolley, C.S., E. Gould, M. Frankfurt, and B.S. McEwen (1990) Naturally occurring fluctuation in dendritic spine density on adult hippocampal pyramidal neurons. *J. Neurosci.* 10:4035–4039.
- Yakovlev, P.I. (1962) Morphological criteria of growth and maturation of the nervous system in man. *Res. Publ. Assoc. Res. Nerv. Ment. Dis.* 39:3–46.
- Yamada, M., Y. Wada, H. Tsukagoshi, E.-I. Otomo, and M. Hayakawa (1988) A quantitative Golgi study of basal dendrites of hippocampal CA1 pyramidal cells in senile dementia of Alzheimer type. *J. Neurol. Neurosurg. Psychiatr.* 51:1088–1090.
- Yoshii, F., W.W. Barker, J.Y. Chang, D. Loewenstein, A. Apicella, D. Smith, T. Boothe, M.D. Ginsberg, S. Pascal, and R. Duara (1988) Sensitivity of cerebral glucose metabolism to age, gender, brain volume, brain atrophy, and cerebrovascular risk factors. *J. Cerebral Blood Flow Metabol.* 8:654–661.
- Yuste, R., and W. Denk (1995) Dendritic spines as basic functional units of neuronal integration. *Nature* 375:682–684.
- Zeki, S.M. (1978) Uniformity and diversity of structure and function in rhesus monkey prestriate visual cortex. *J. Physiol.* 277:273–290.
- Zilles, K., R. Werners, U. Büsching, and A. Schleicher (1986) Ontogenesis of the laminar structure in areas 17 and 18 of the human visual cortex. A quantitative study. *Anat. Embryol. (Berlin)* 174:339–353.